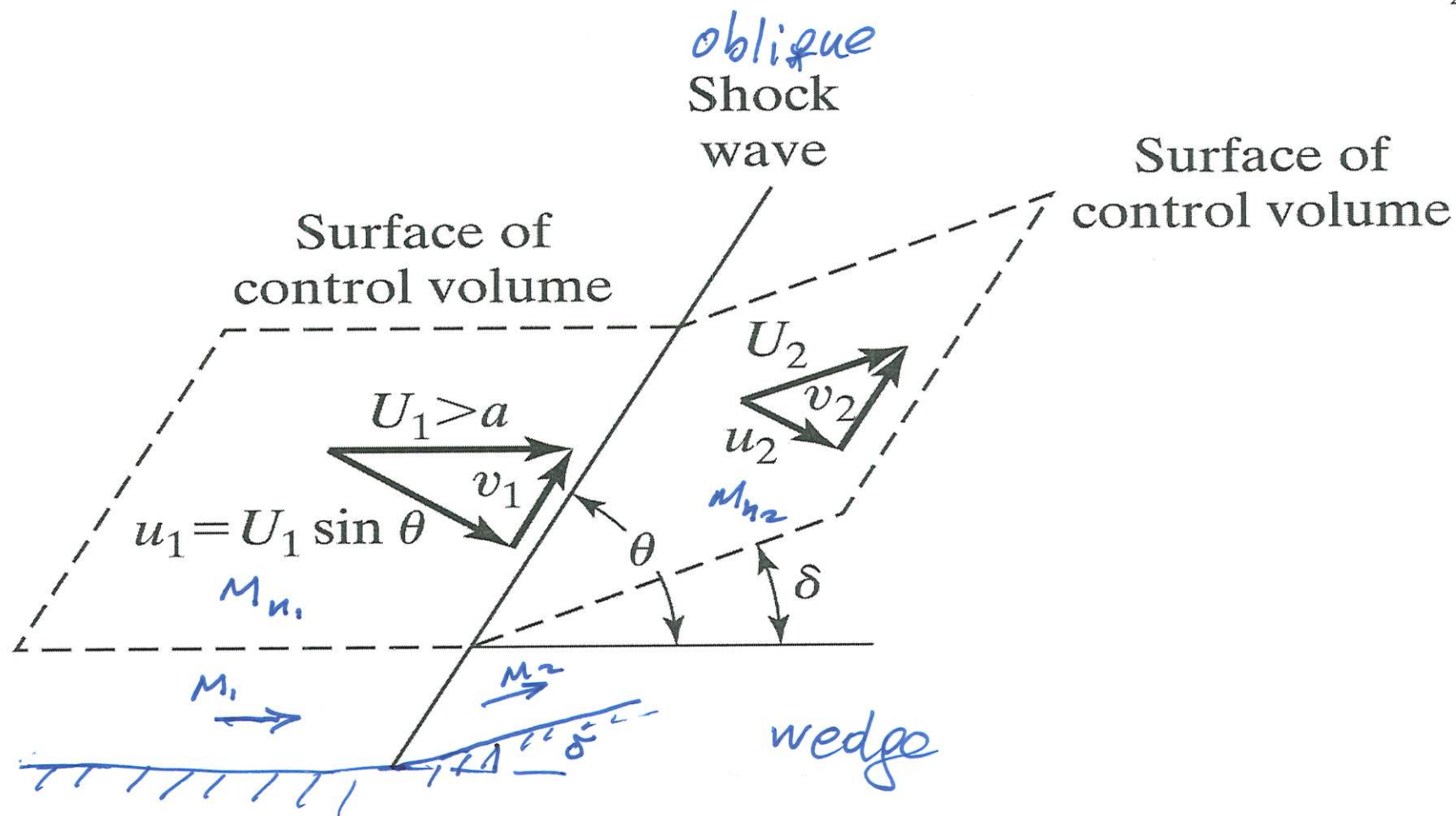


Normal and Oblique Shocks [8.6-8.10]

Objectives:

- Develop relations for normal and oblique shocks
- Study viscous effects, shock boundary layer interactions
- Introduction to shock/shock interactions
- Introduction to supersonic tunnel testing



Consider control volume around the shock

let $u_1, u_2 \Rightarrow$ velocity components $\perp$ the shock

$v_1, v_2 \Rightarrow$ velocity components $\parallel$ to the shock

continuity $\rho_1 u_1 = \rho_2 u_2$ (8.64)

tangential momentum $\rho_1 u_1 v_1 = \rho_2 u_2 v_2$ (8.66)

(same tangential components) $\Rightarrow \underline{v_1 = v_2}$ (8.70)

normal momentum:

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \quad (8.65)$$

energy equation (for adiabatic flow)

$$h_1 + \frac{U_1^2}{2} = h_2 + \frac{U_2^2}{2} \quad (8.67)$$

$$(U_1^2 = u_1^2 + v_1^2)$$

$$\vec{U}^2 = u^2 + v^2$$

$$\Rightarrow h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \quad (8.71)$$

Also, perfect gas $p = \rho RT$ (1.10)

Enthalpy *equation*: $h = c_p T$ (8.15)

Normal component of velocity is used above:

$$u_1 = U_1 \sin \theta \Rightarrow M_{n_1} = M_1 \sin \theta \quad (8.72)$$

System of 5 eqs with 5 unknowns can be solved to find qualities behind the shock (using M_{n1})

M- δ - θ relation (need 2 out of 3)

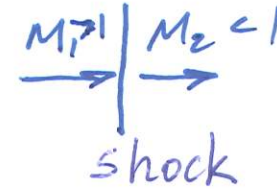
$$\cot \delta = \tan \theta \left[\frac{(\gamma + 1)M_1^2}{2(M_{n_1}^2 - 1)} - 1 \right] \quad (8.73)$$

// $M_1 \sin \theta$

- normal shocks are a special case of oblique shocks for

$$\theta = \frac{\pi}{2}, \quad \delta = 0$$

(no deflection)



$$\frac{p_2}{p_1} = \frac{2\gamma M_{n_1}^2 - (\gamma - 1)}{\gamma + 1} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n_1}^2 - 1) \quad (8.74)$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_{n_1}^2}{(\gamma - 1)M_{n_1}^2 + 2} \quad (8.75)$$

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \frac{\rho_1}{\rho_2} \Rightarrow \frac{T_2}{T_1} = \frac{[2\gamma M_{n_1}^2 - (\gamma - 1)][(\gamma - 1)M_{n_1}^2 + 2]}{(\gamma + 1)^2 M_{n_1}^2} \quad (8.76)$$

$$M_{n_2}^2 = \frac{M_{n_1}^2 + [2 / (\gamma - 1)]}{[2\gamma / (\gamma - 1)] M_{n_1}^2 - 1} = \frac{1 + [(\gamma - 1) / 2] M_{n_1}^2}{\gamma M_{n_1}^2 - (\gamma - 1) / 2}$$

The Mach number behind the shock can be found from

$$M_2 = \frac{M_{n_2}}{\sin(\theta - \delta)} \Rightarrow M_2^2 = \frac{1 + [(\gamma - 1) / 2] M_{n_1}^2}{[\gamma M_{n_1}^2 - (\gamma - 1) / 2] \sin^2(\theta - \delta)}$$

$$\frac{p_{t_2}}{p_{t_1}} = e^{-(s_2 - s_1)/R} = \left[\frac{(\gamma + 1) M_{n_1}^2}{(\gamma - 1) M_{n_1}^2 + 2} \right]^{\gamma/(\gamma-1)} \left[\frac{(\gamma + 1)}{2\gamma M_{n_1}^2 - (\gamma - 1)} \right]^{1/(\gamma-1)} \quad (8.78)$$

$$p_{t_2} < p_{t_1}$$

Please note the change in entropy

$$s_2 - s_1 < 0 \quad \text{for} \quad M_{n_1} < 1 \quad \text{entropy cannot reduce (not physical)}$$

$$s_2 - s_1 = 0 \quad \text{for} \quad M_{n_1} = 1 \quad (\text{sound/Mach wave})$$

$$s_2 - s_1 > 0 \quad \text{for} \quad M_{n_1} > 1 \quad (\text{shock wave})$$

$$C_p = \frac{p_2 - p_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_1^2}$$

Trends and graphs are below

TABLE 8.4 Flow Property Changes across Weak Shock Waves

<i>Normal shock</i>	<i>Oblique shock</i>
$p_2 > p_1$	$p_2 > p_1$
$\rho_2 > \rho_1$	$\rho_2 > \rho_1$
$T_2 > T_1$	$T_2 > T_1$
$M_2 < 1$	$M_2 < M_1$
$p_{t,2} < p_{t,1}$	$p_{t,2} < p_{t,1}$
$T_{t,2} = T_{t,1}$	$T_{t,2} = T_{t,1}$

total pressure decreases (adiabatic)

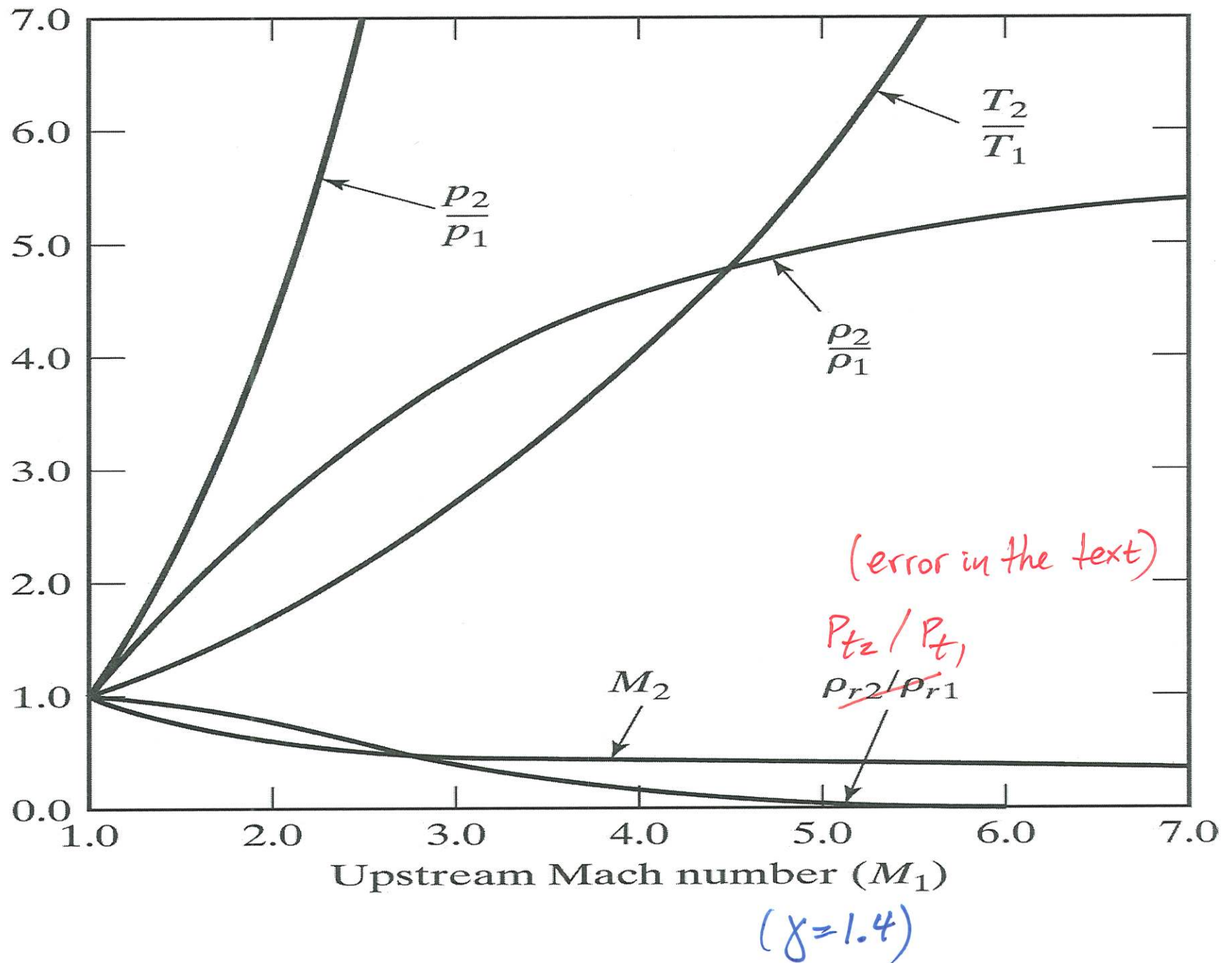
limits

for $M_1 \rightarrow \infty$ $M_2 \rightarrow \sqrt{\frac{\gamma-1}{2\gamma}} = 0.378$

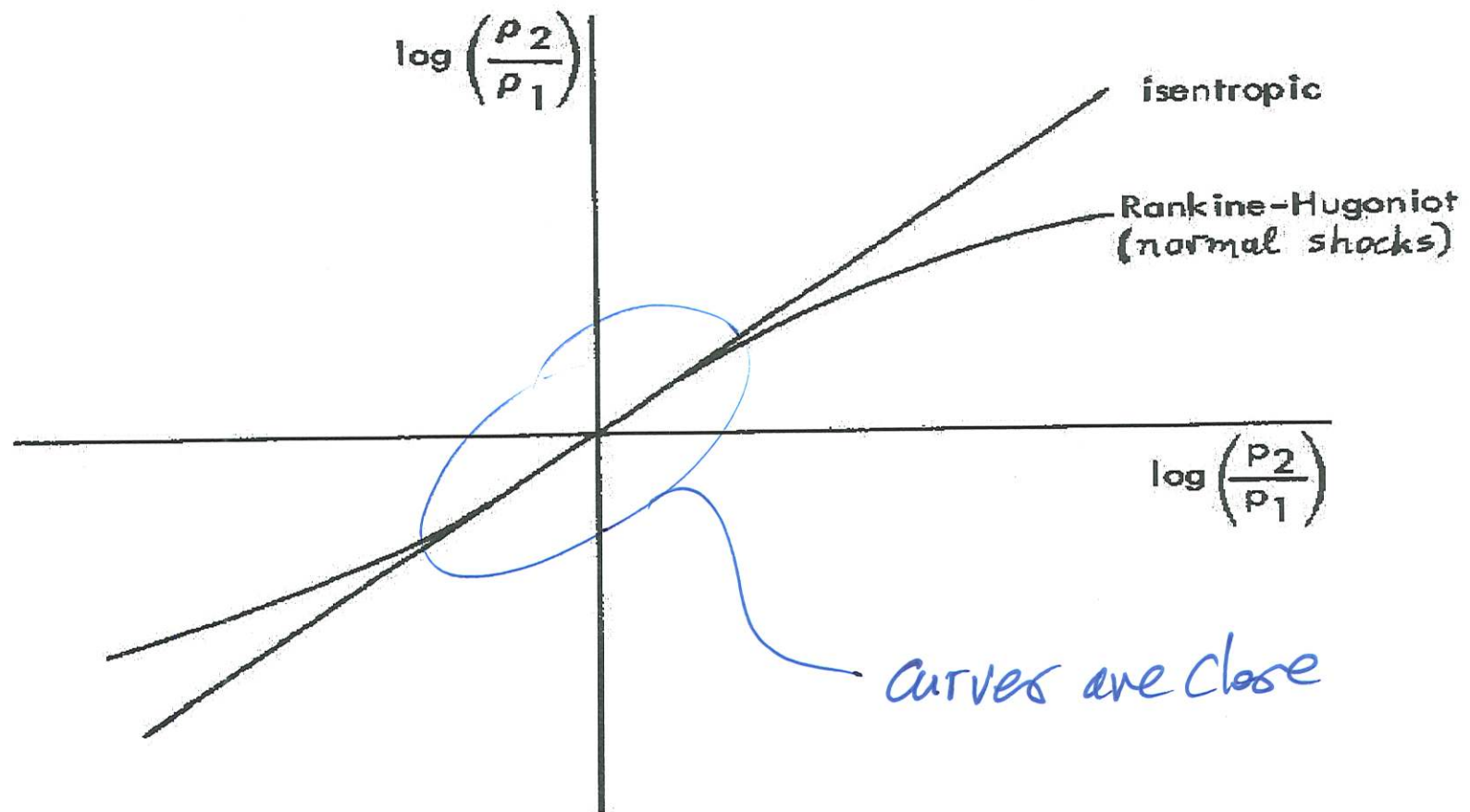
(strength) $\frac{p_2}{p_1} \rightarrow \infty$

$$\frac{\rho_2}{\rho_1} \Rightarrow \frac{\gamma+1}{\gamma-1} = 6$$

$$\frac{T_2}{T_1} \rightarrow \infty$$

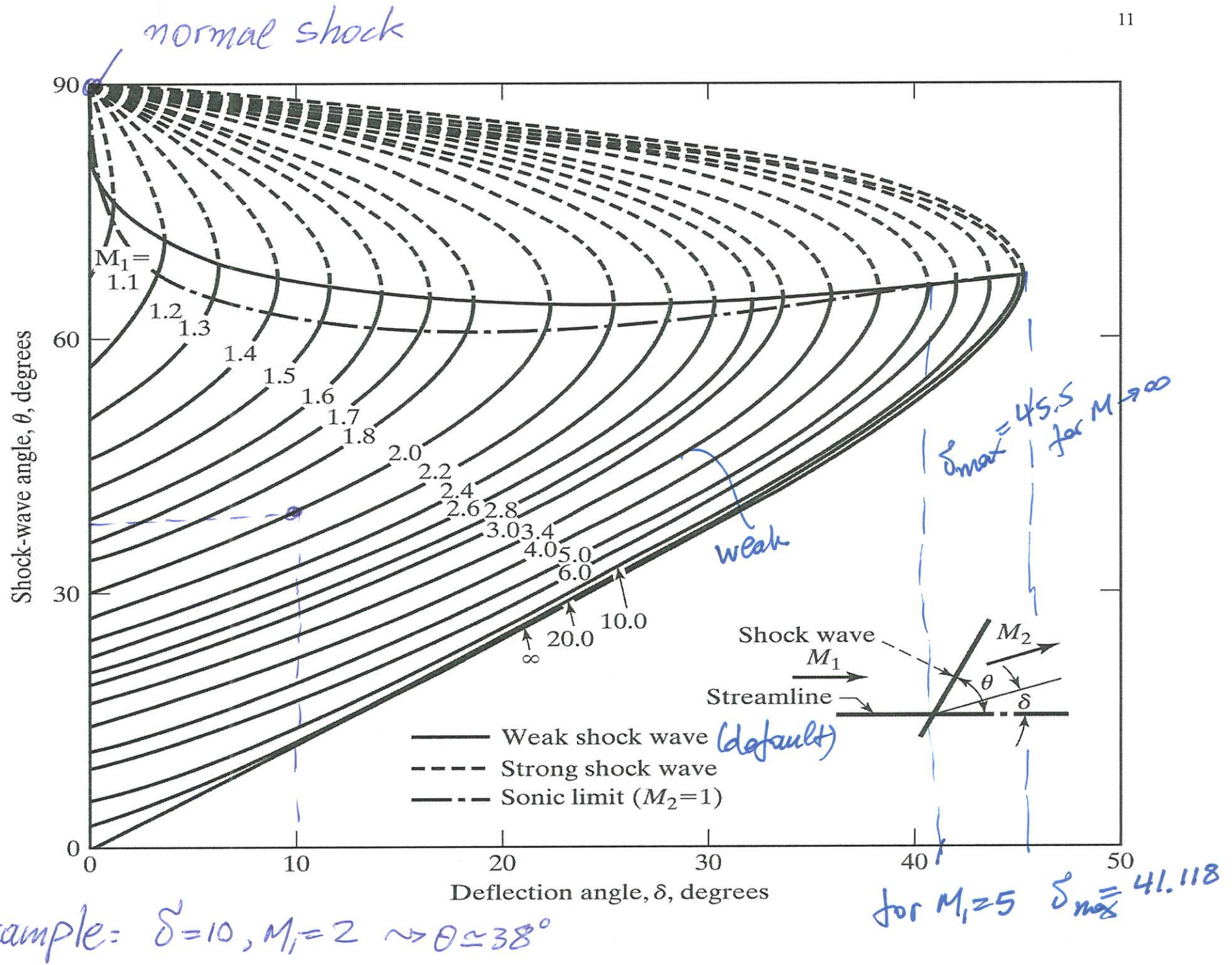


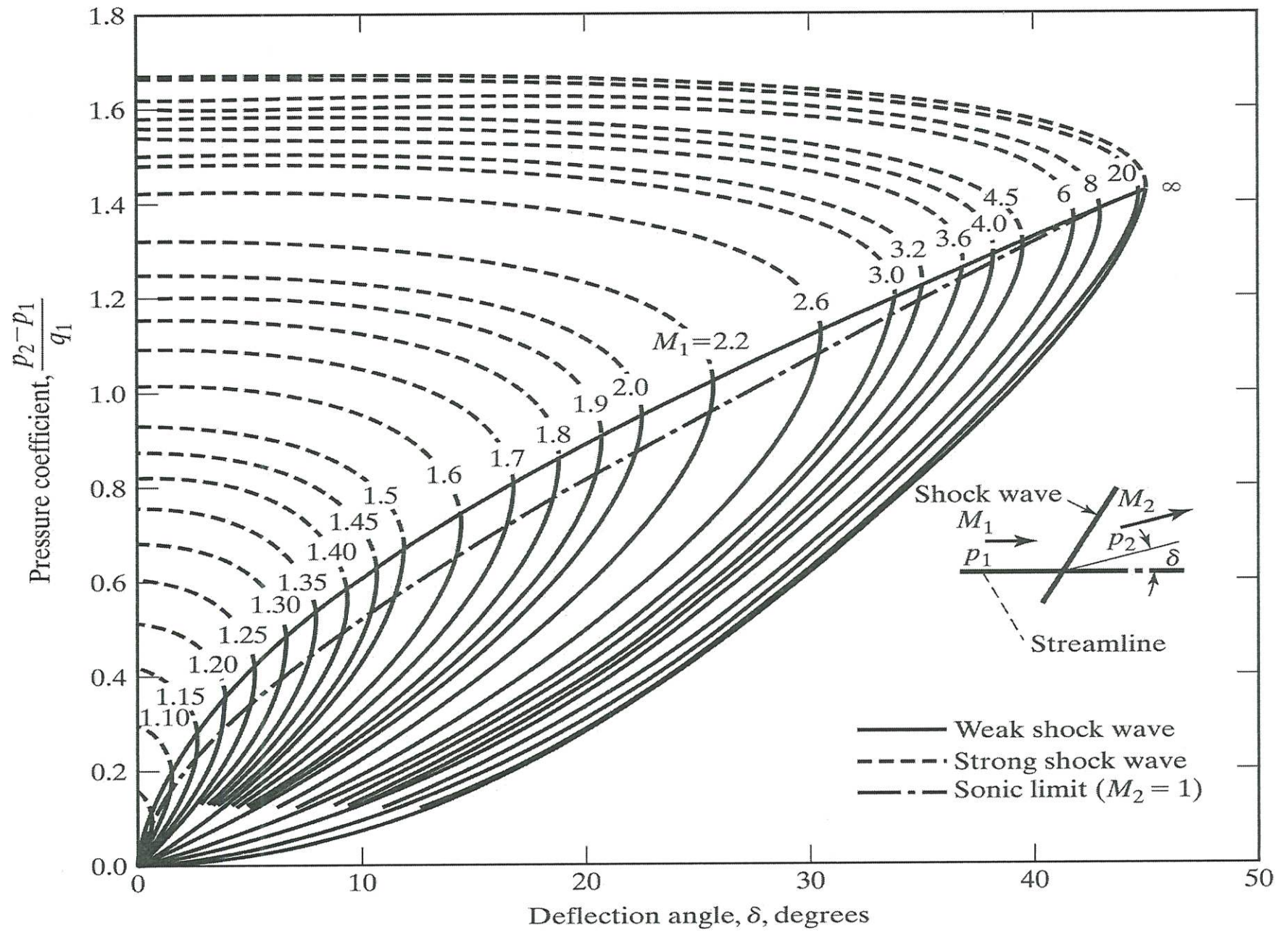
Comparison between isentropic and normal shock relations

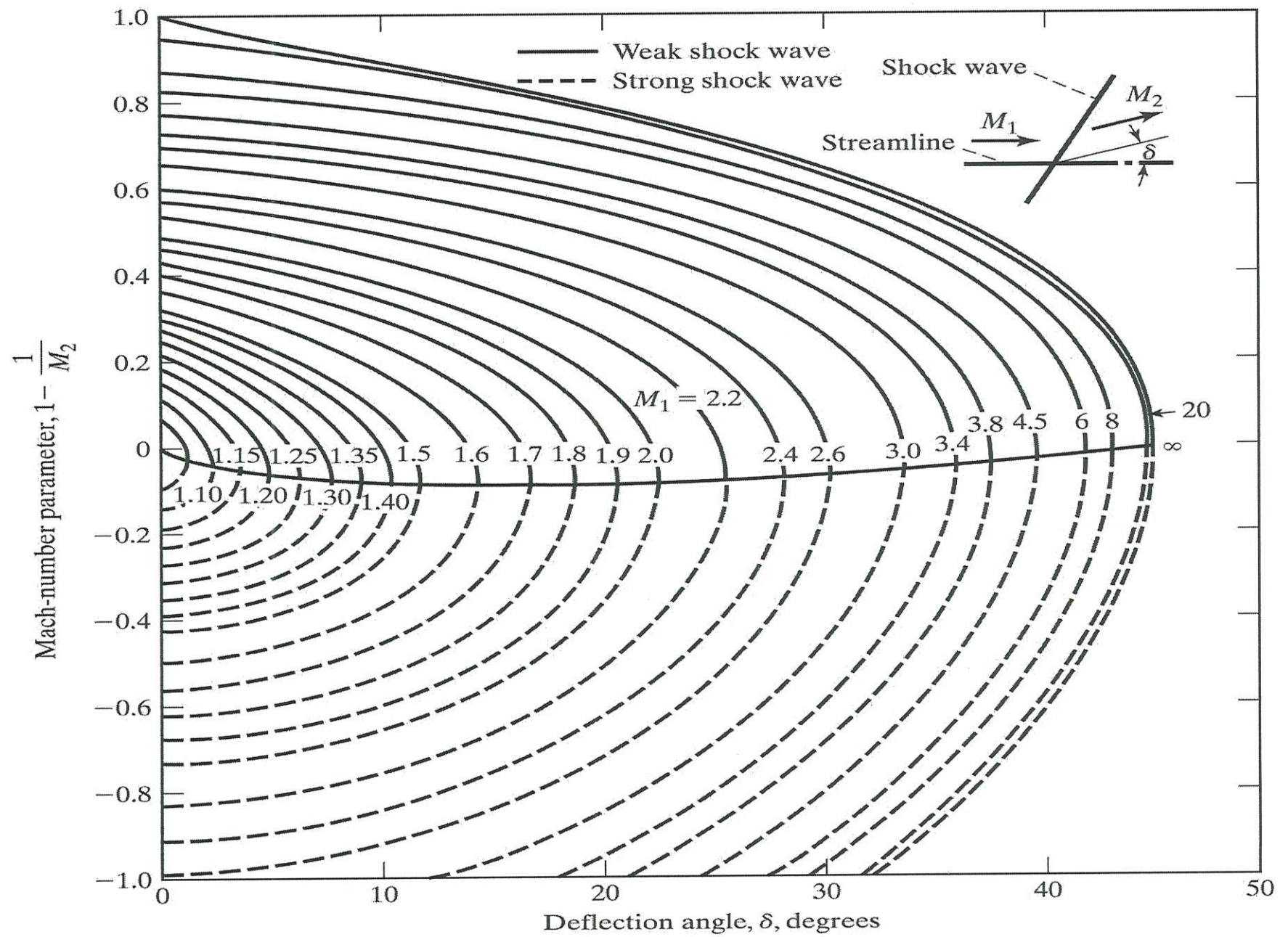


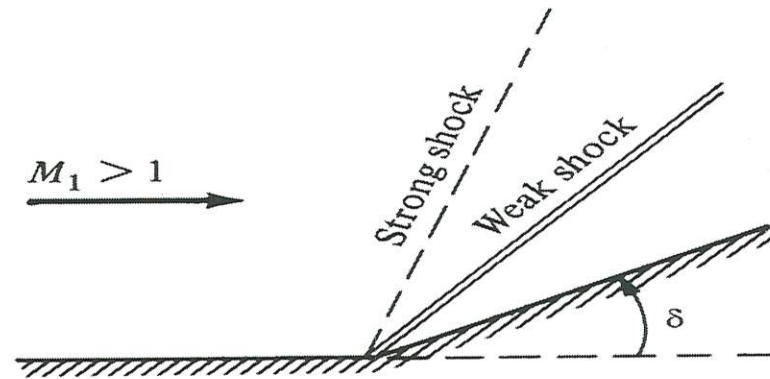
$$(M=1.2 \div 1.3)$$

Weak shocks can be assumed isentropic.

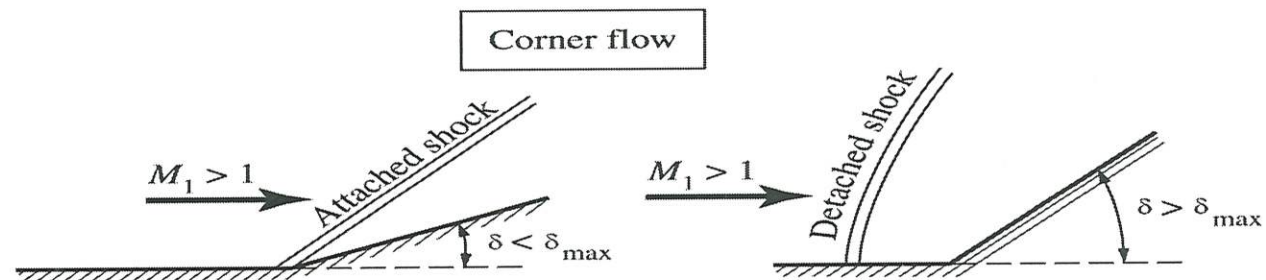
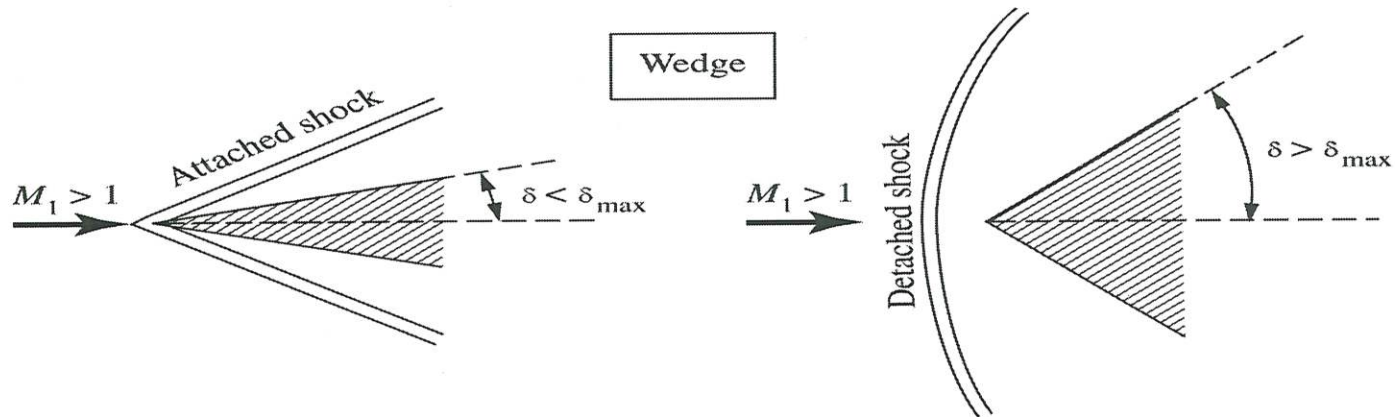








strong shock
has a higher θ



- For any given M_1 there is a $\delta_{\max}$ (see graphs)

$\delta > \delta_{\max}$ detached shock

$\delta < \delta_{\max}$ 2 values of θ

- large (strong shock)
- small (weak shock)

Back pressure determines the solution;

weak shock usually occurs (default)

use unless specified otherwise

for a *strong* shock $M_2 < 1$

for a *weak* shock $M_2 > 1$ (except near $\theta_{\max}$)

$$\delta = 0 \Rightarrow \theta = \frac{\pi}{2} \quad (\text{strong}) \quad \text{normal shock}$$

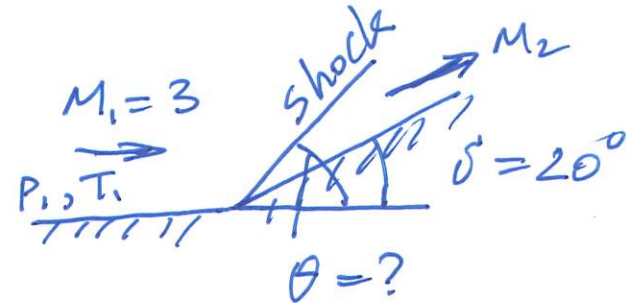
$$\rightarrow \theta = \mu \quad (\text{weak}) \quad \text{Mach wave}$$

- for a fixed δ as $M_1 \downarrow$ then $\theta \uparrow$ till the shock detaches.
(weak solution)
- as $M_{n1} \uparrow \Rightarrow$ stronger shock $\frac{P_2}{P_1} \uparrow$
- as $\delta \uparrow$ (with M_1 constant) $\Rightarrow$ stronger shock, $\theta \uparrow$ (weak)
- as $M_1 \uparrow$ (with δ constant) $\Rightarrow$ stronger shock, $\theta \downarrow$ (weak)

example

$$M_1=3, p_1 = 1 \text{ atm}, T_1 = 288\text{k}, \delta = 20^\circ$$

find $p_2, T_2, M_2, p_{t2}, T_{t2}$



from $\theta - \delta - M$ diagram: $M_1 = 3$
 $\delta = 20^\circ \Rightarrow \theta = 37.5^\circ$ (weak)

$$\Rightarrow M_{n_1} = M_1 \sin \theta = 1.826 \quad (>1)$$

from normal shock tables, for $M_{n_1} = 1.826$

$$M_{n_1} = 1.826 \Rightarrow \frac{p_2}{p_1} = 3.723 \quad ; \quad \frac{T_2}{T_1} = 1.551$$

$$M_{n_2} = 0.6108$$

$$p_{t_2}/p_{t_1} = 0.8011$$

$$\Rightarrow p_2 = \underset{\text{atm}}{3.723}, \quad T_2 = 446.7^{\text{ok}}, \quad M_2 = \frac{M_{n_2}}{\sin(\theta - \delta)} = 2.03 \quad (>1)$$

from isentropic tables

use M_1, M_2
not M_{n1}, M_{n2}

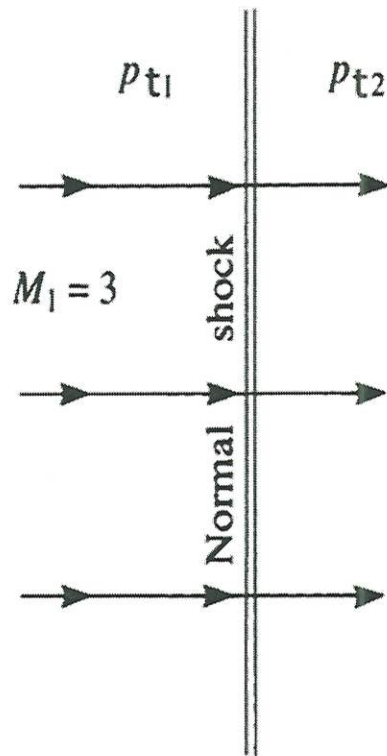
$$M_2 < M_1$$

$$M_1 = 3 \Rightarrow \frac{p_{t1}}{p_1} = 36.73, \quad \frac{T_{t1}}{T_1} = 2.8$$

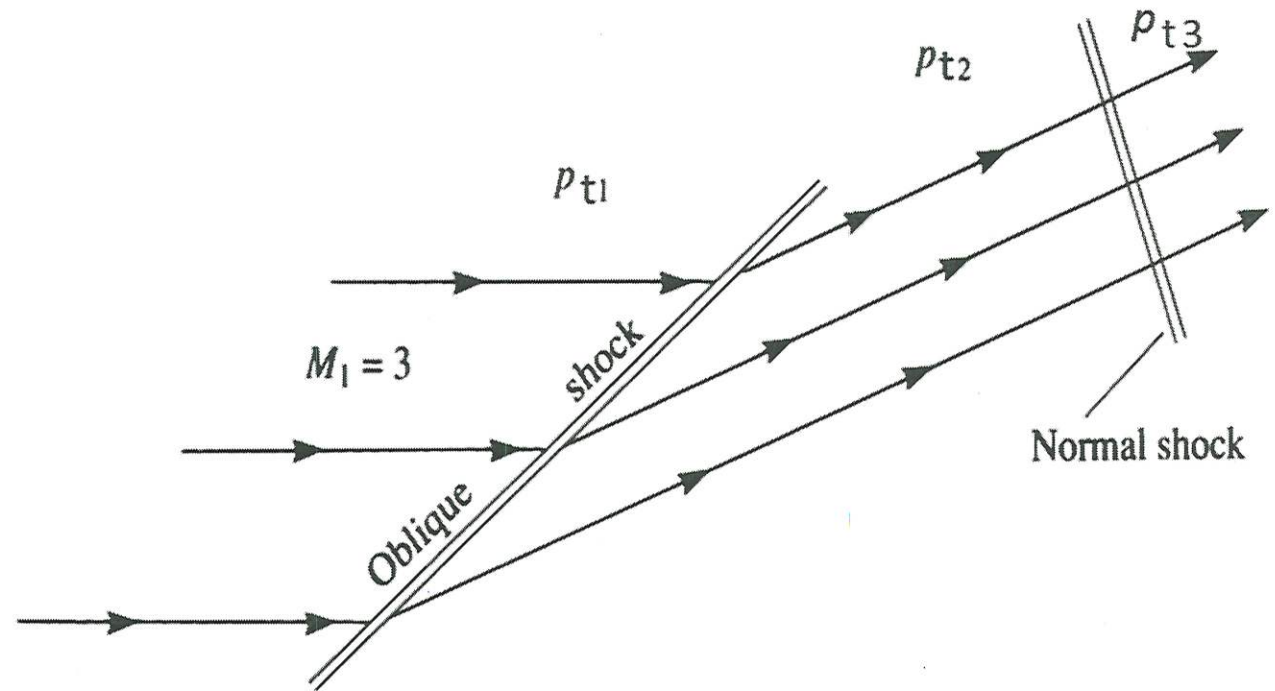
$$T_{t2} = T_{t1} = \frac{T_{t1}}{T_1} T_1 = 806.4$$

adiabatic

$$\begin{aligned} \text{or } \frac{p_{t2}}{p_2} &= \frac{p_{t2}}{p_2} \cdot p_2 = 8.2 \times 3.723 \quad \left(\text{for } M_2 = 2.03 \right) \\ p_{t2} &= \frac{p_{t2}}{p_{t1}} \cdot \frac{p_{t1}}{p_1} \cdot p_1 = 29.42 \text{ atm} \\ &\quad \left(\frac{0.8011}{\text{from normal shock tables}} \right) \left(\frac{36.73}{\text{for } M_1 = 3} \right) \end{aligned}$$



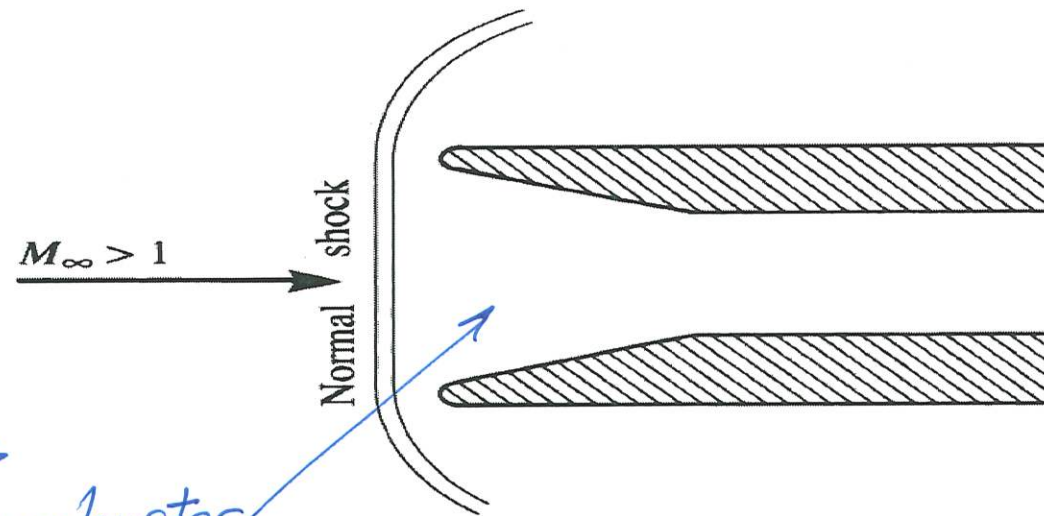
Case 1



Case 2

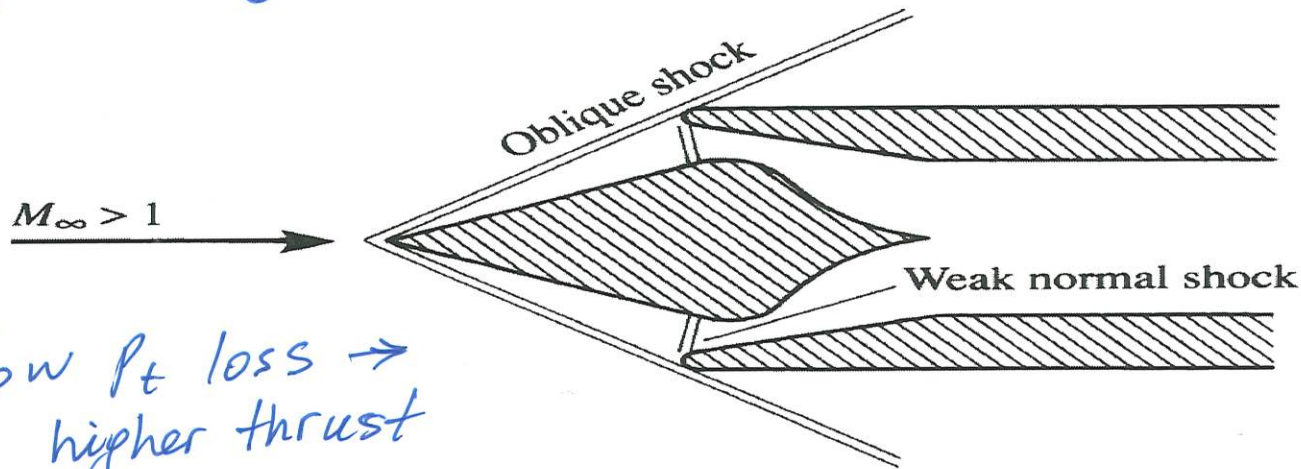
Final total pressure is 76% higher for case 2 (for $\theta = 40^\circ$)

[do it as an exercise]



high P_t loss
flow entering combustor
high P , high T

(a) Normal shock inlet (undesirable)



low P_t loss $\rightarrow$
higher thrust

scramjet

(b) Oblique shock inlet (desirable)

$$C_p = \frac{p - p_\infty}{q_\infty}$$

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 = \frac{1}{2} \frac{\gamma p_\infty}{\gamma p_\infty} \rho_\infty U_\infty^2$$

$\frac{1}{a_\infty^2}$

$$= \frac{\gamma p_\infty U_\infty^2}{2 a_\infty^2} = \frac{\gamma}{2} p_\infty M_\infty^2$$

$$\Rightarrow C_p = \frac{p - p_\infty}{\frac{\gamma}{2} p_\infty M_\infty^2} \Rightarrow \boxed{C_p = \frac{2}{\gamma M_\infty^2} \left(\frac{p}{p_\infty} - 1 \right)}$$

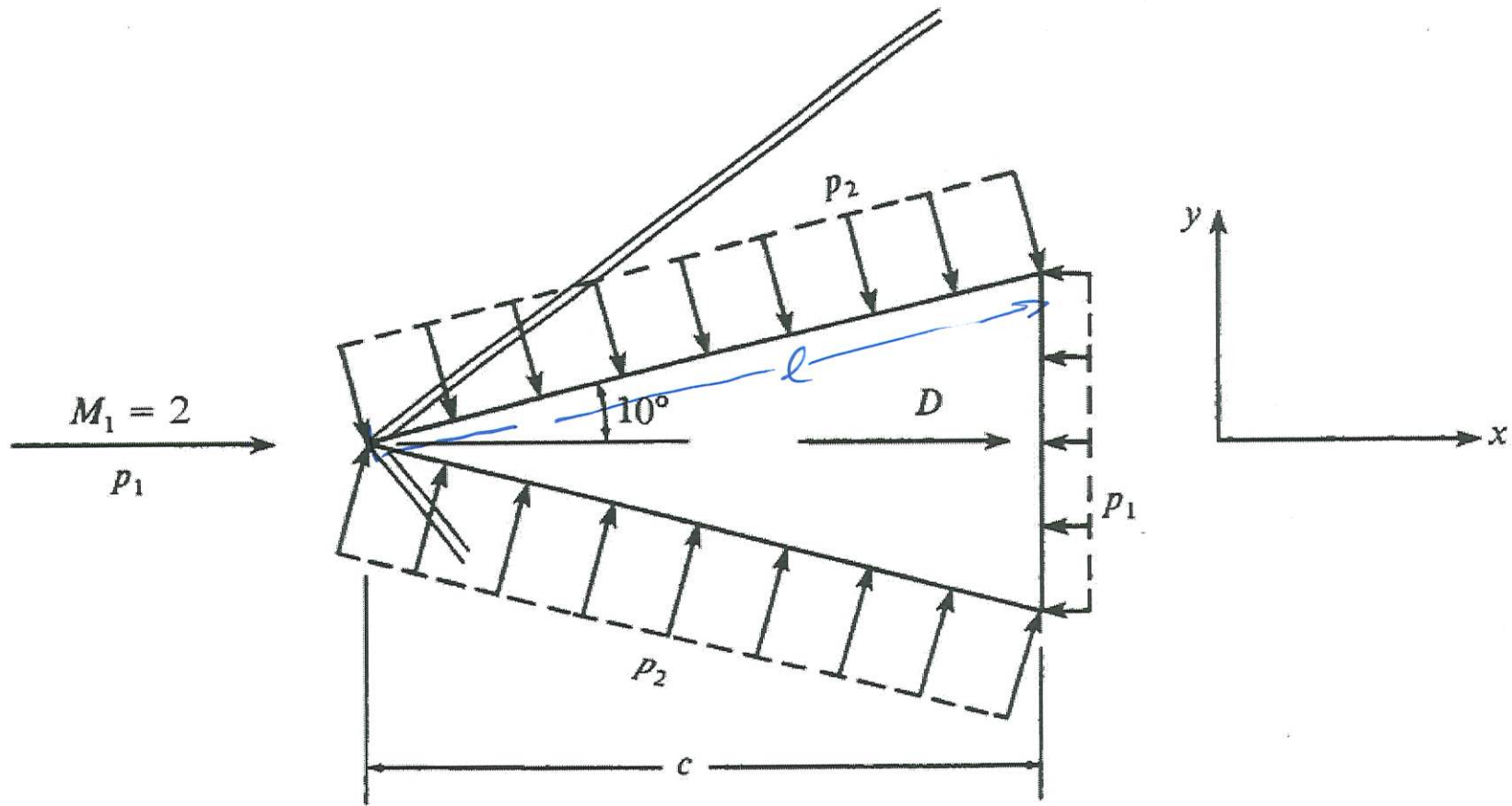
[∞] for 1 and 2 conditions: ^{nothing}

$$C_p = \frac{2}{\gamma M_1^2} \left(\frac{p_2}{p_1} - 1 \right)$$

³⁰⁷ for incompressible flow:

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 \quad \left(\text{using Bernoulli's eq.} \right)$$

Consider a 10° wedge, $M_1=2$ find C_p , C_d

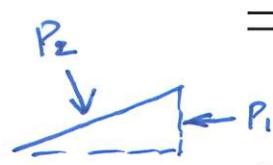


$M_1 = 2$, $\delta = 10^\circ$ from $\theta - \delta - M$ diagram $\theta = 39.314^\circ$

$$\Rightarrow M_{n_1} = M_1 \sin \theta = 2 \times \sin 39.314 = 1.267$$

from normal shock tables:

$$M_{n_1} = 1.267 \Rightarrow \frac{P_2}{P_1} = 1.707$$



$$\Rightarrow C_p = \frac{P_2 - P_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_1^2} = \frac{4(1.267^2 - 1)}{2.4 \times 2^2} = 0.252$$

same answer

also: $C_p = \frac{2}{\gamma M_1^2} \left(\frac{P_2}{P_1} - 1 \right)$

$$D' = 2 p_2 l \sin \delta - 2 p_1 l \sin \delta = 2(\sin \delta) l (P_2 - P_1)$$

/ 2 oblique sides vertical side

- horizontal component

drag/unit span

- (or use gauge pressure)

$l = c / \cos \delta$

$$= 2 c \tan \delta (P_2 - P_1)$$

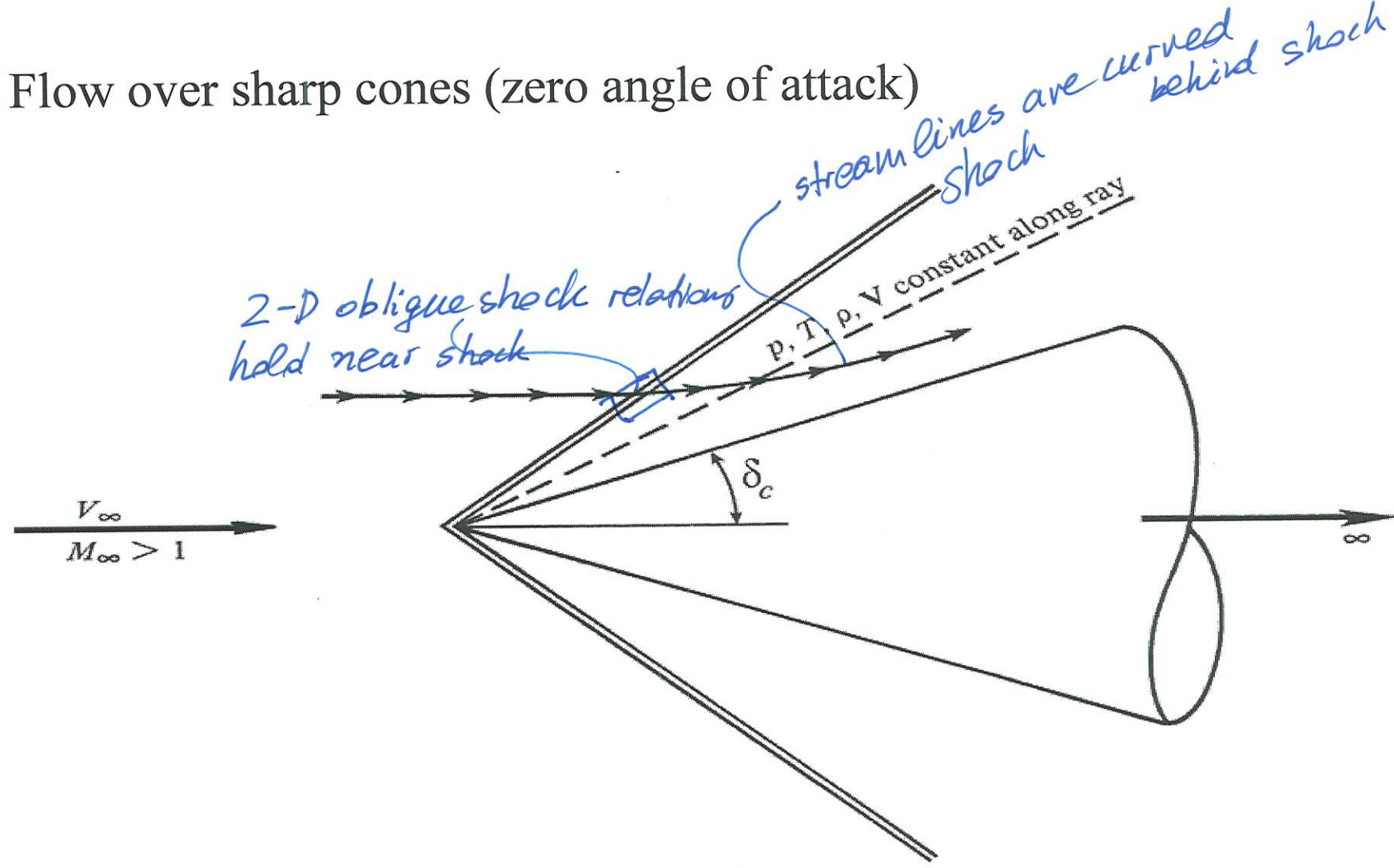
$$C_d = \frac{D'}{q_1 c} = \frac{D'}{\frac{\gamma}{2} P_1 M_1^2 c} = \frac{4 \tan \delta}{\gamma M_1^2} \left(\frac{P_2}{P_1} - 1 \right) = 2 \tan \delta C_p$$

$$= 2 \times \tan 10^\circ \times 0.252$$

$$= 0.184$$

note $C_d \neq 0$; oblique shock creates drag

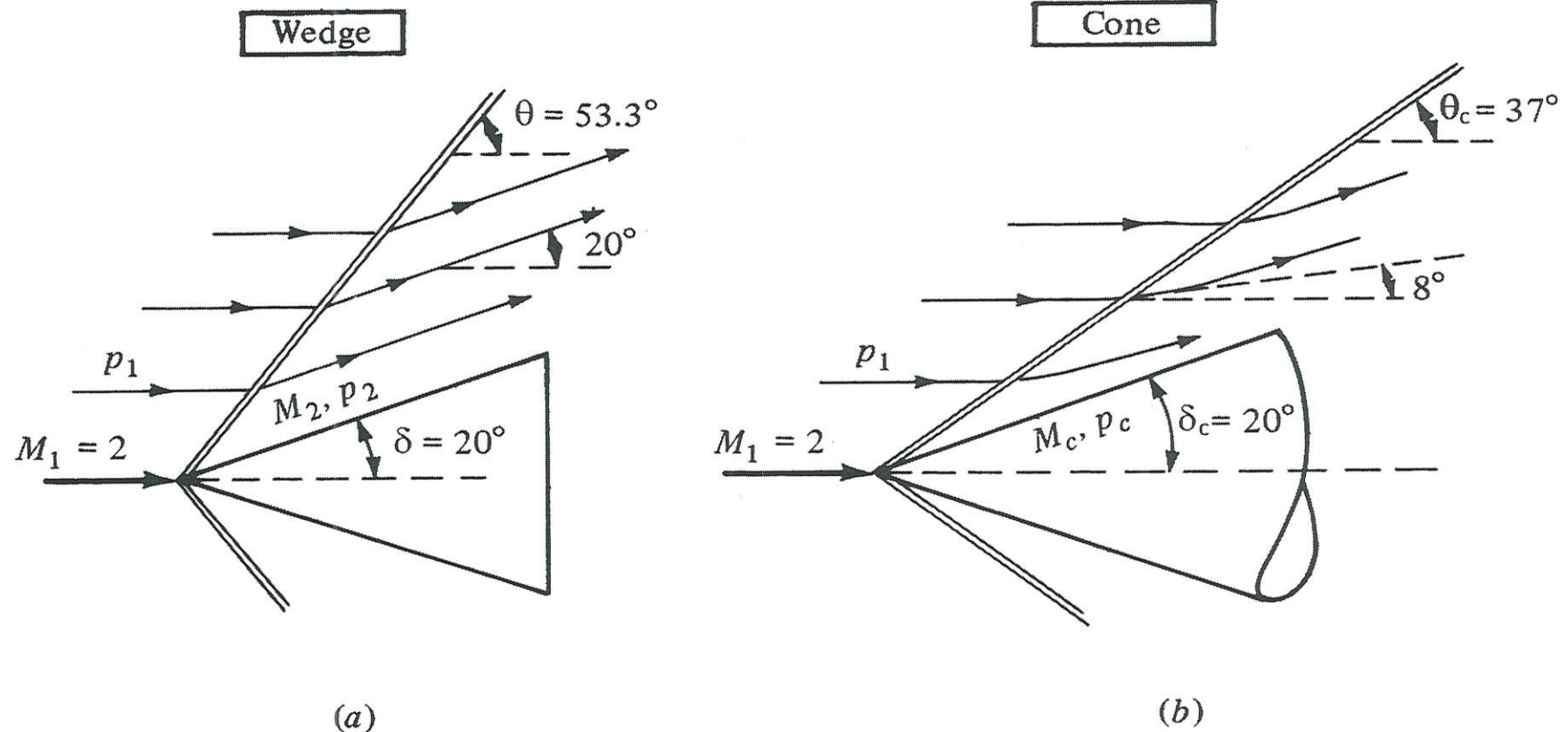
Flow over sharp cones (zero angle of attack)



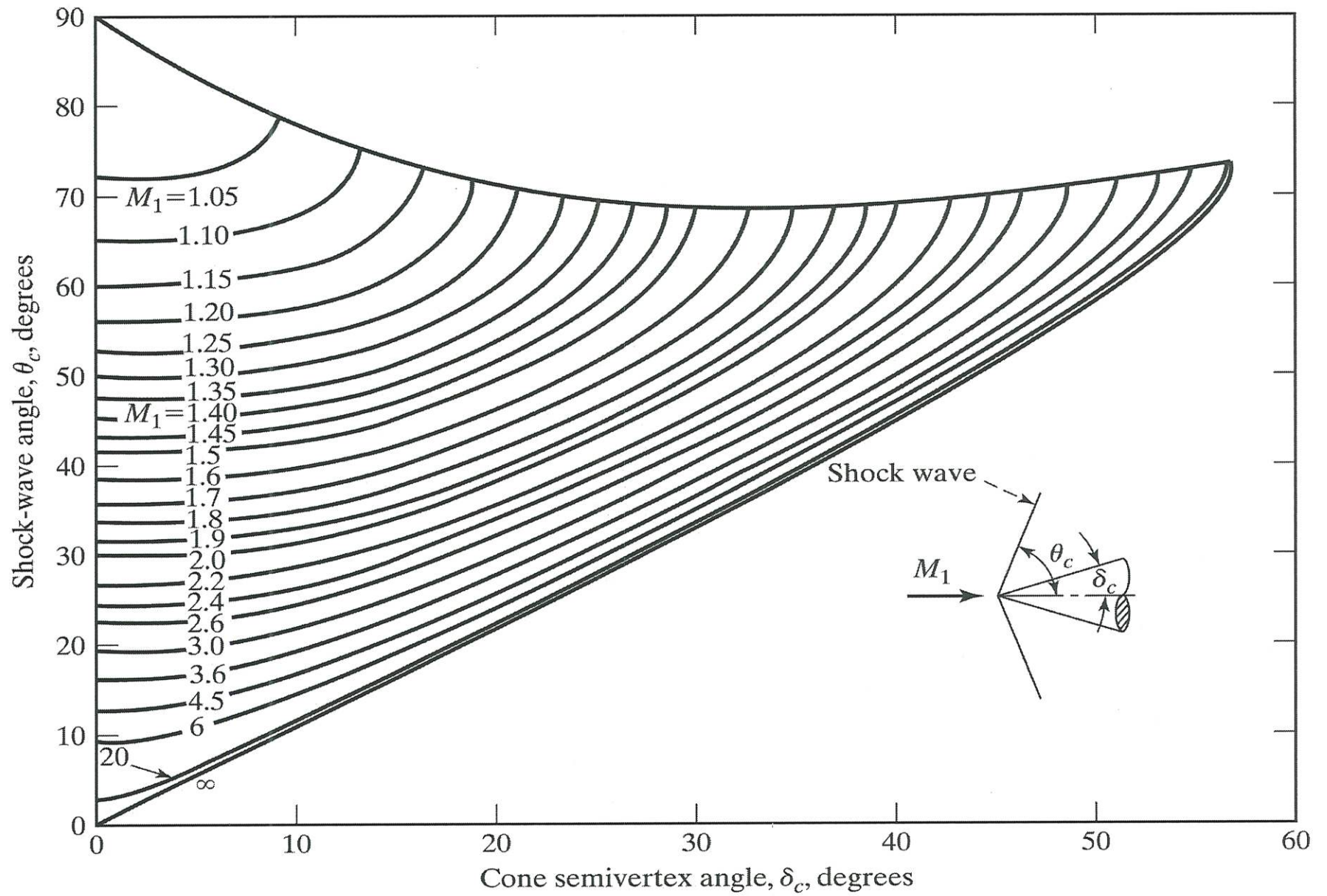
$$C_p = \frac{p_c - p_1}{q_1}$$

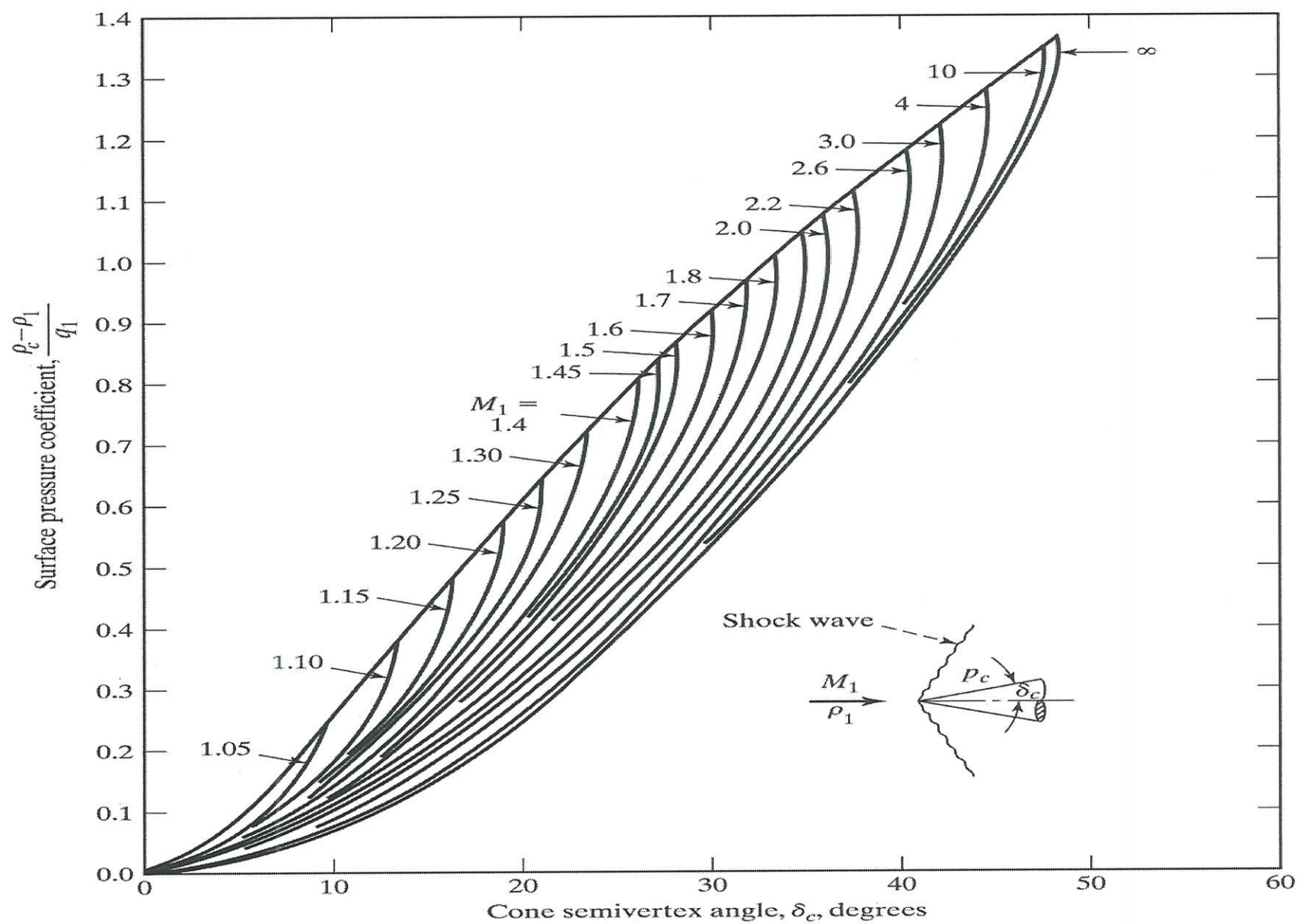
p_c : pressure on the cone surface
(constant along the surface)

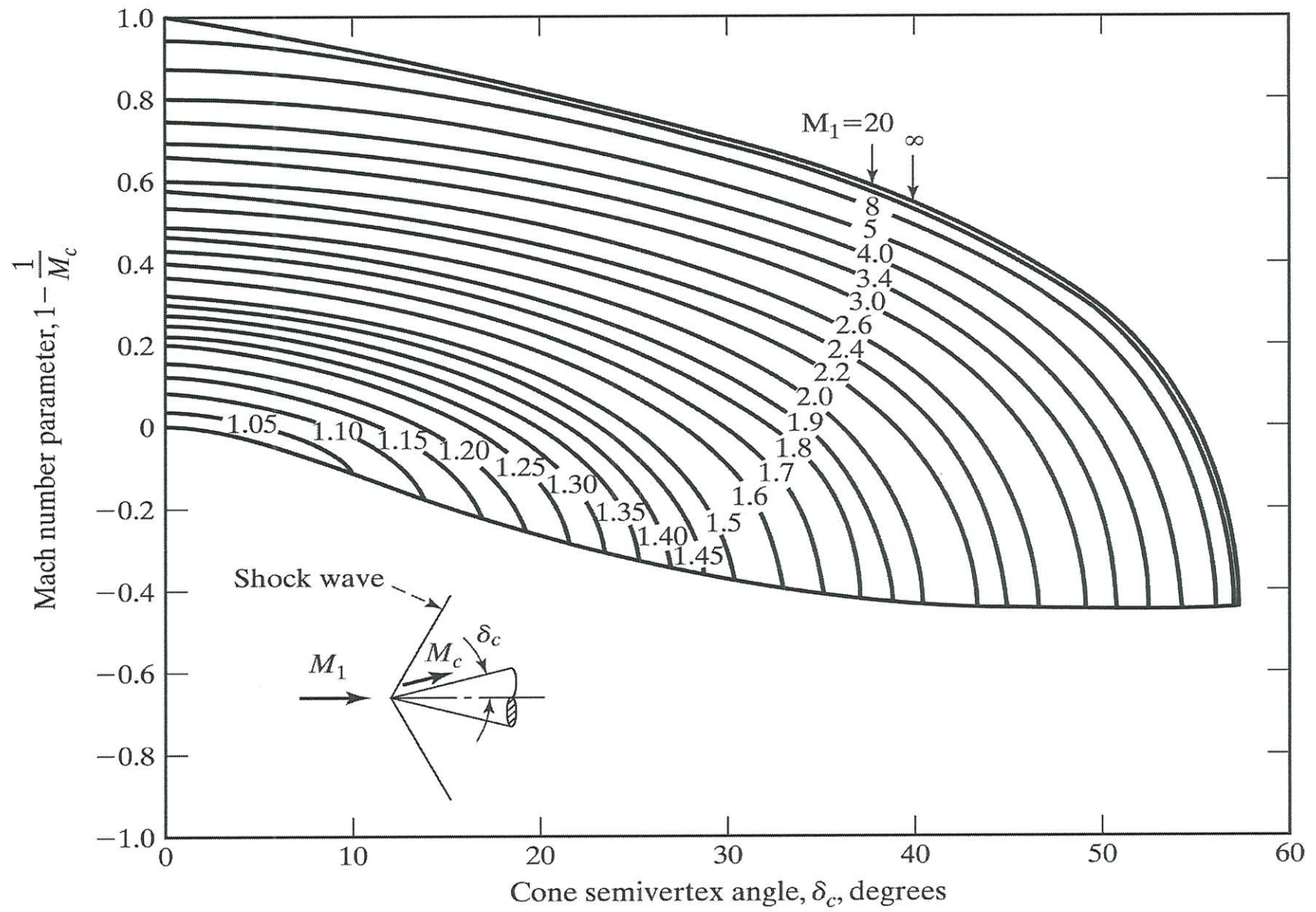
Compare wedges – cones



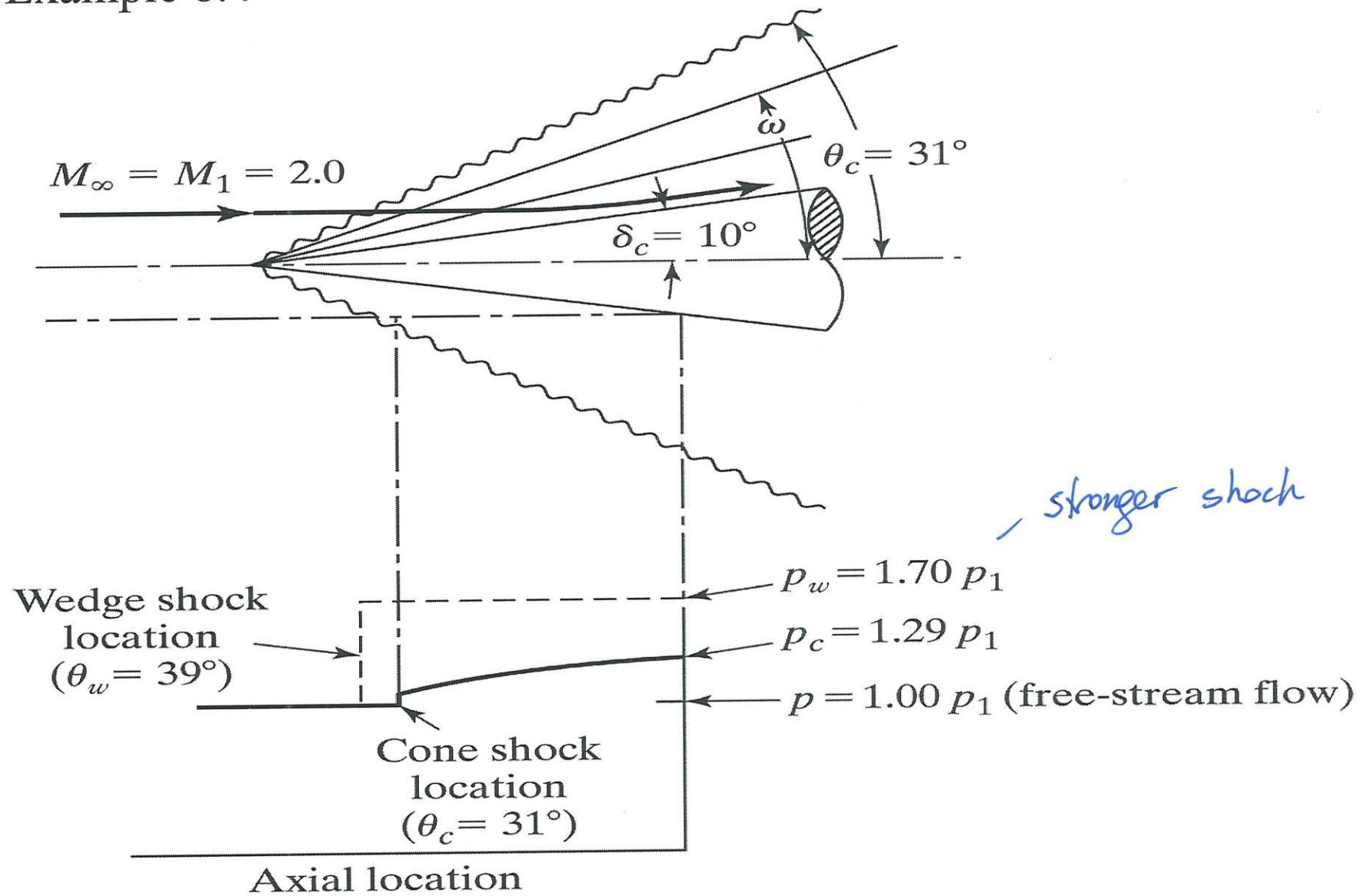
θ is smaller for a cone; 3rd dimension gives extra space for the flow to move around object $\Rightarrow$ 3-D relieving effect
(similar to flow over cylinder / sphere)







Example 8.4



Semi-vertex angle $\delta_c = 10^\circ$ $M_1 = 2$

Fig 8.16a (cone graphs) $\theta_c = 31^\circ$

Shock tables (2D) or (8.74) for $M_1 = 2$ $\theta_c = 31^\circ$ $p_2/p_1 = 1.07$

On the cone surface: $p_c/p_1 = 1.29$

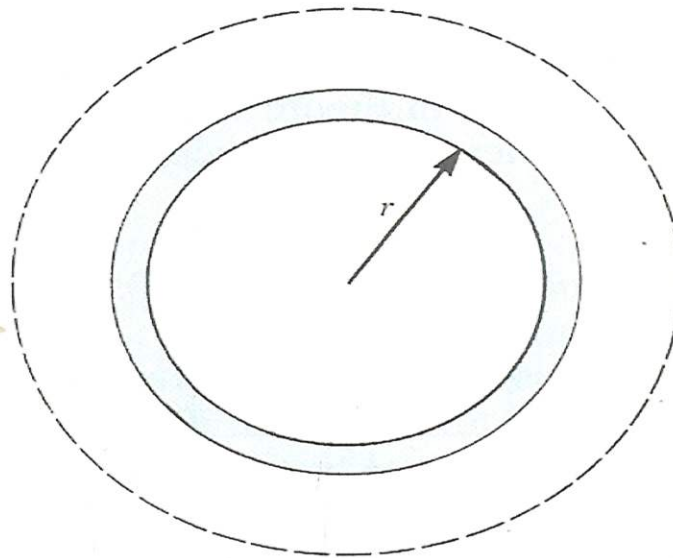
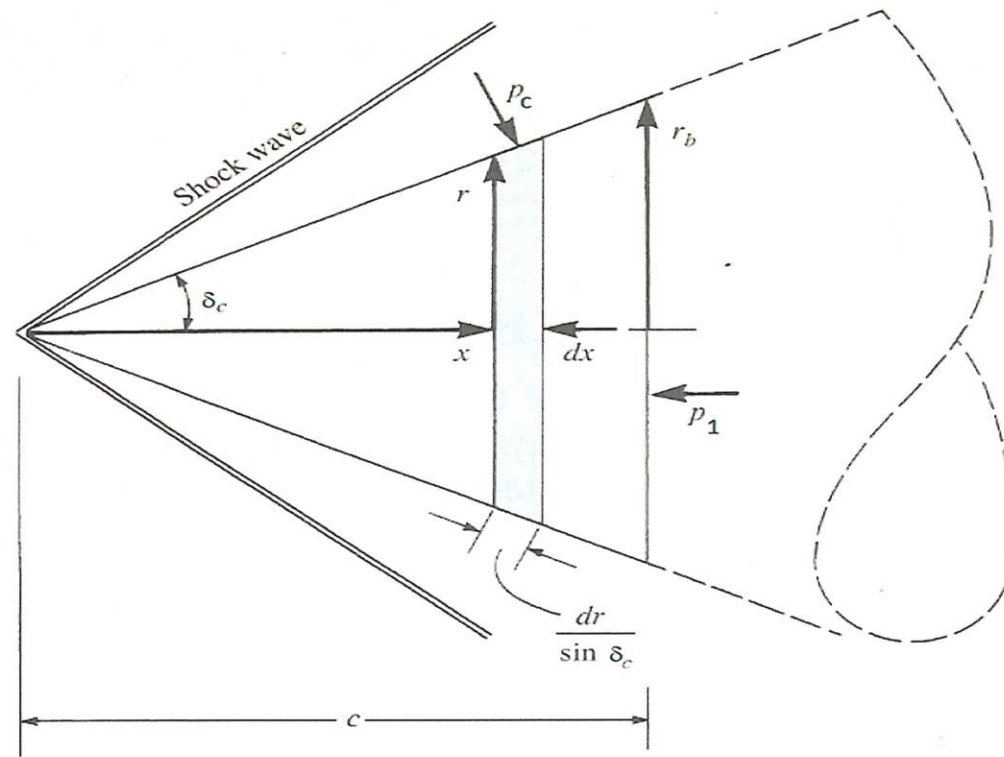
$$C_p = \frac{p_c - p_1}{q_1} = \frac{2}{\gamma M_1^2} \left(\frac{p_c}{p_1} - 1 \right) = \frac{2}{1.4 \times 2^2} (1.29 - 1) = 0.104$$

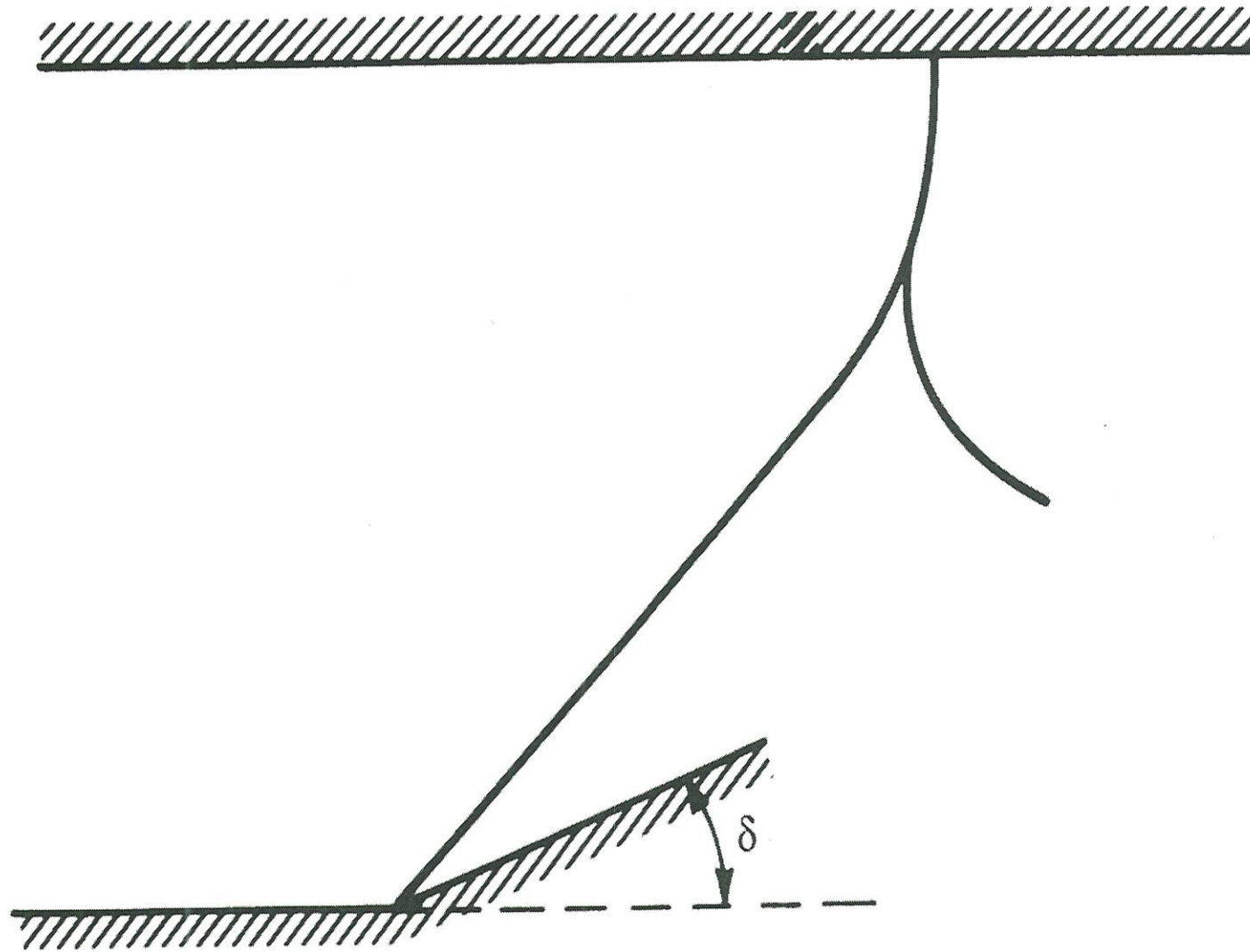
$$D = \int_0^{r_b} 2\pi r p_c dr - \int_0^{r_b} r 2\pi p_1 dr = \int_0^{r_b} 2\pi r (p_c - p_1) dr = \pi (p_c - p_1) r_b^2$$

oblique surface *base*
horizontal force

$$C_D = \frac{D}{q_1 \pi r_b^2} = \frac{\pi r_b^2 (p_c - p_1)}{\pi r_b^2 q_1} = C_p \quad (C_D = C_p)$$

$= 0.104$





If δ increases (or M_1 decreases) a reflected shock might not be possible;

Detached shock wave in front of blunt body

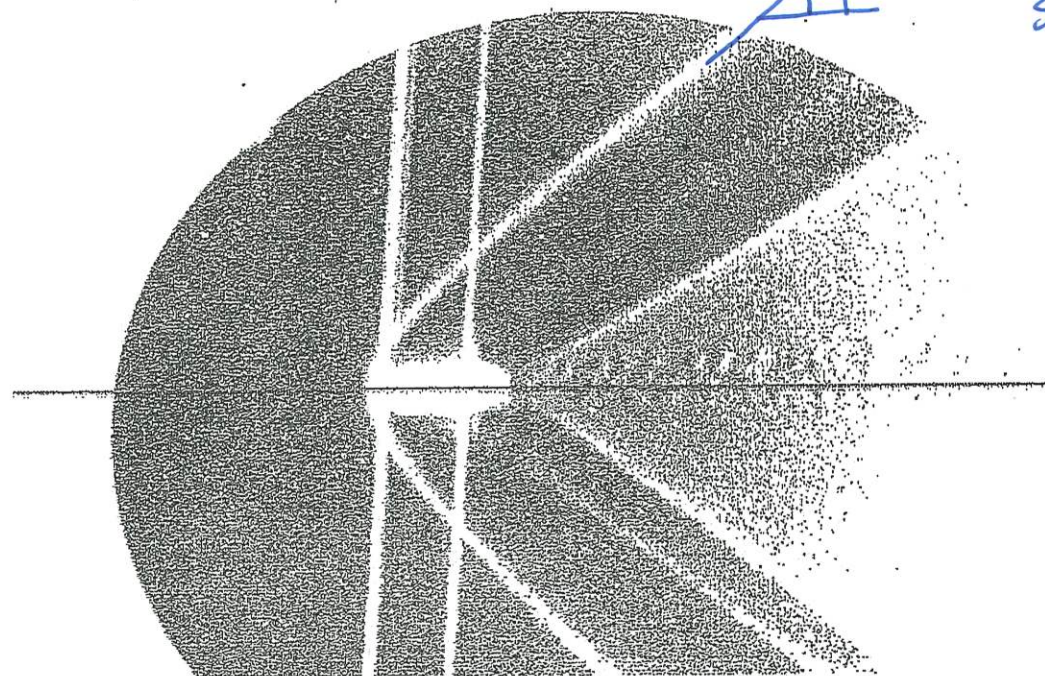
- a. normal
- b. oblique shock (strong)
- c. dividing point between strong-weak solution
- d. flow supersonic behind the shock
- e. weak solution
- f. Mach wave (far-field solution)

shape of shock and detachment distance δ can be found only from numerical solutions.

The first image of supersonic flow was taken by Ernst Mach and is shown below. It has been first published in 1887 and is often reproduced in textbooks without specifying the free-stream Mach number (since the concept was not introduced until 1920s).

Based on the information in the figure, estimate the free-stream Mach number for this flow.

(Note: two parallel vertical lines are due to trip wires and can be ignored).

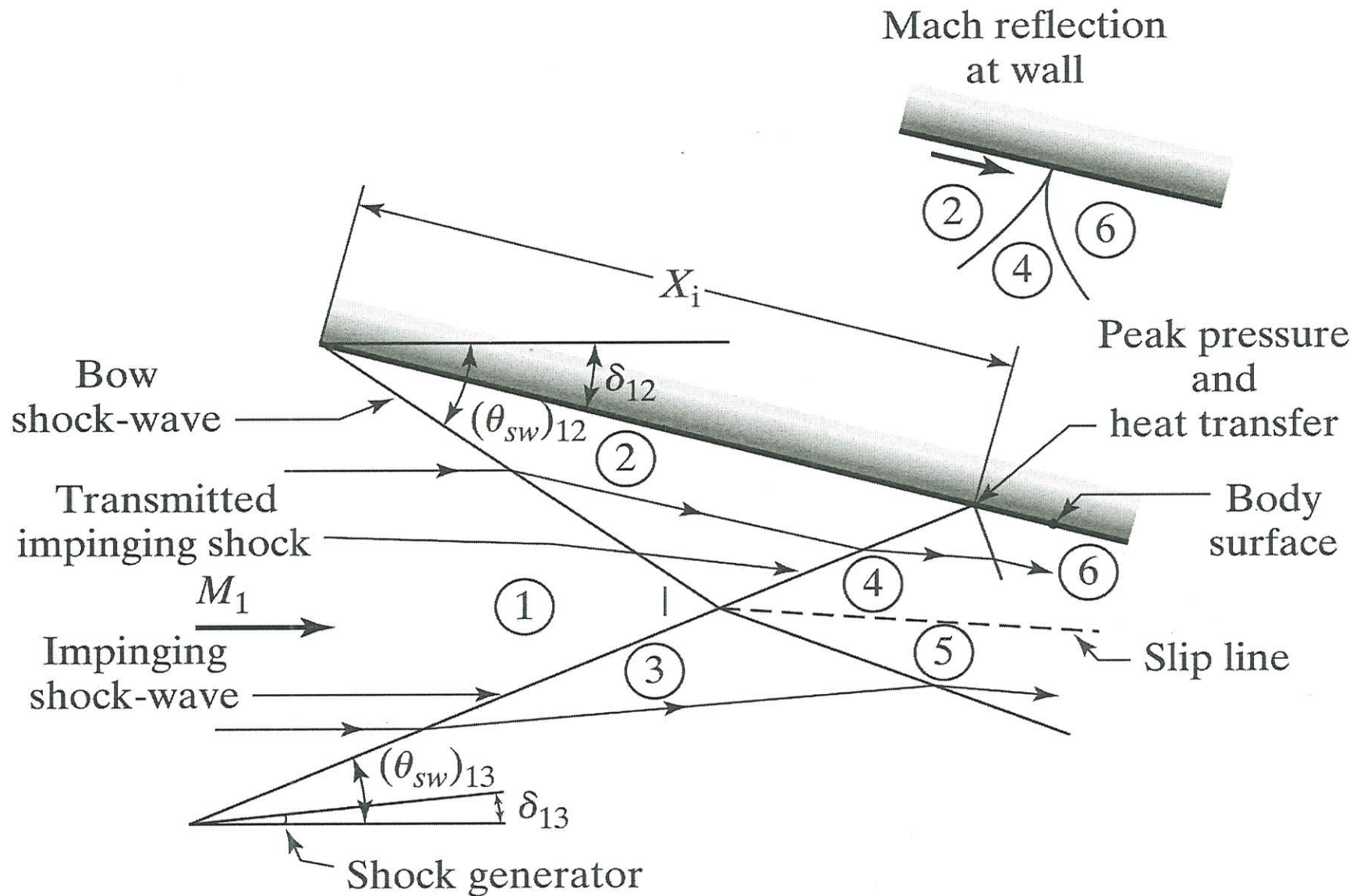


$$\sin \mu = \frac{1}{M}$$

$$\sim 45^\circ$$

$$\Rightarrow M \approx 1.4$$

Shock/Shock interactions (8.9)



between sections 4 & 5

Slip line: entropy changes, $p_4 = p_5$ same pressure.

Velocities in the same direction, magnitude different

Example 8.6: $M_1=6$, $p_1=10^{-3}$ atm, $\delta_{12}=15^\circ$, $\delta_{13}=5^\circ$

Region 2: $M_2=3.992$; $p_2 / p_1 = 6.073$; $p_2 = 6.073 \times 10^{-3}$ atm

Region 3: $M_3=5.316$; $p_3 / p_1 = 2.010$; $p_3 = 2.010 \times 10^{-3}$ atm
(flow angle wrt condition ①)

Iterations for θ_f (to get $p_4 = p_5$); start with $\theta_f = -10$ ($= \delta_{13} - \delta_{12}$)

θ_f	δ_{24}	p_4 / p_2	$p_4 = \left(\frac{p_4}{p_2} \cdot \frac{p_2}{p_1} \cdot p_1 \right)$	θ_{35}	p_5 / p_3	$p_5 = \left(\frac{p_5}{p_3} \cdot \frac{p_3}{p_1} \cdot p_1 \right)$
-10	5	1.619	9.832×10^{-3}	15	5.167	10.386×10^{-3}

[$p_4 < p_5 \rightarrow$ decrease $|\theta_f|$]

-9.5	5.5	1.694	10.287	14.5	4.946	9.941
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[$p_4 > p_5 \rightarrow$ increase $|\theta_f|$]

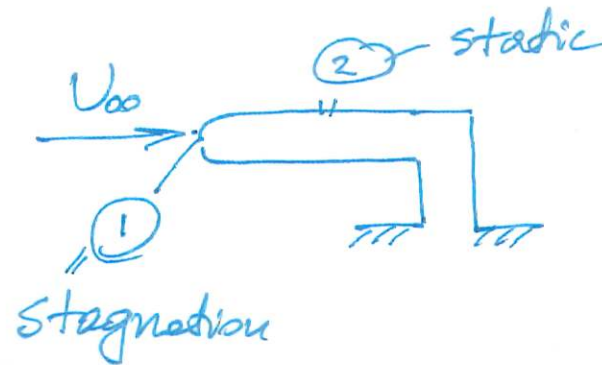
-9.7	5.3	1.663	10.099	14.7	5.034	10.118
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[$p_4 \approx p_5$ close enough]

Velocity Measurement

(Pitot-static tube)

a. Incompressible (201)

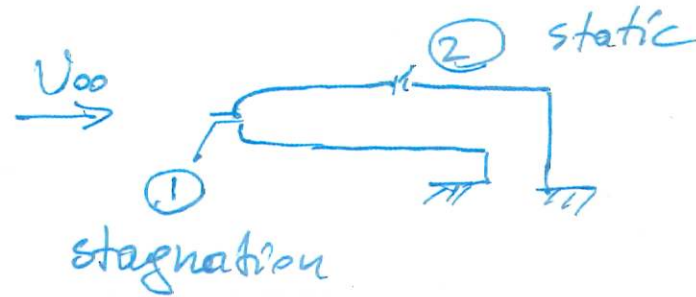


Bernoulli's equation: $p_1 + \frac{1}{2} \cancel{\rho u_1^2} = p_2 + \frac{1}{2} \rho u_2^2 \rightarrow u_2 = U_\infty$
 $u_1 = 0$

$$\Rightarrow U_\infty = \sqrt{\frac{2}{\rho}(p_1 - p_2)}$$

(need to know ρ)

b. Compressible subsonic



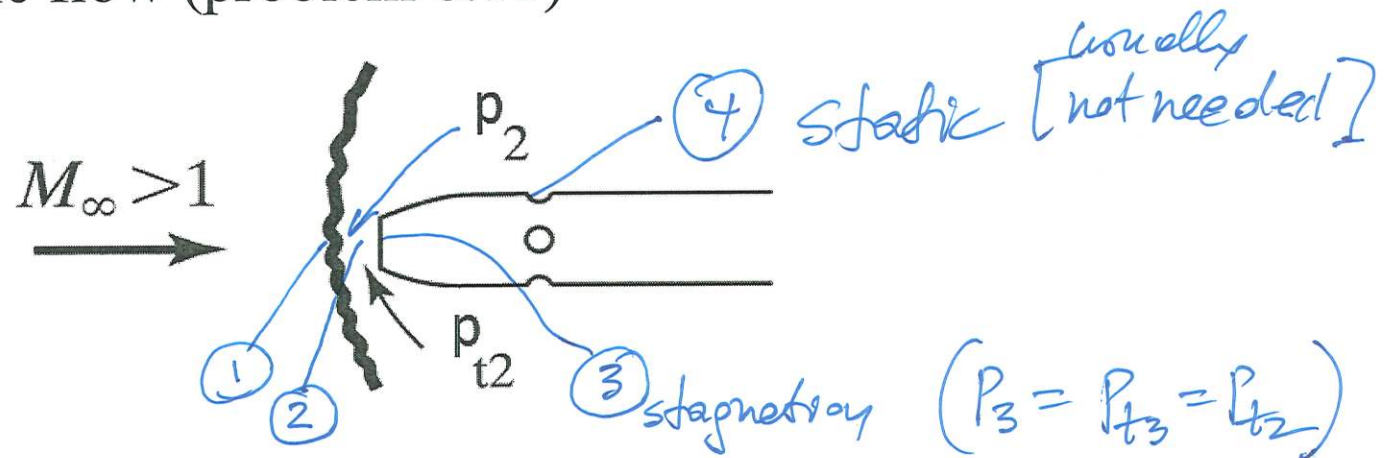
Assume isentropic flow: ^(8.51)

$$\frac{p_t}{p} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\gamma/(\gamma - 1)}$$

$p_t = p_1$
 $p = p_2$
 $M = M_{\infty}$

$$\Rightarrow M_{\infty}^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_1}{p_2} \right)^{(\gamma - 1)/\gamma} - 1 \right]$$

b. Supersonic flow (problem 8.11)



Isentropic
$$\frac{p_{t2}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2 \right)^{\gamma(\gamma-1)} \quad (8.36)$$

Across shock
$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1) \quad (8.75)$$

normal

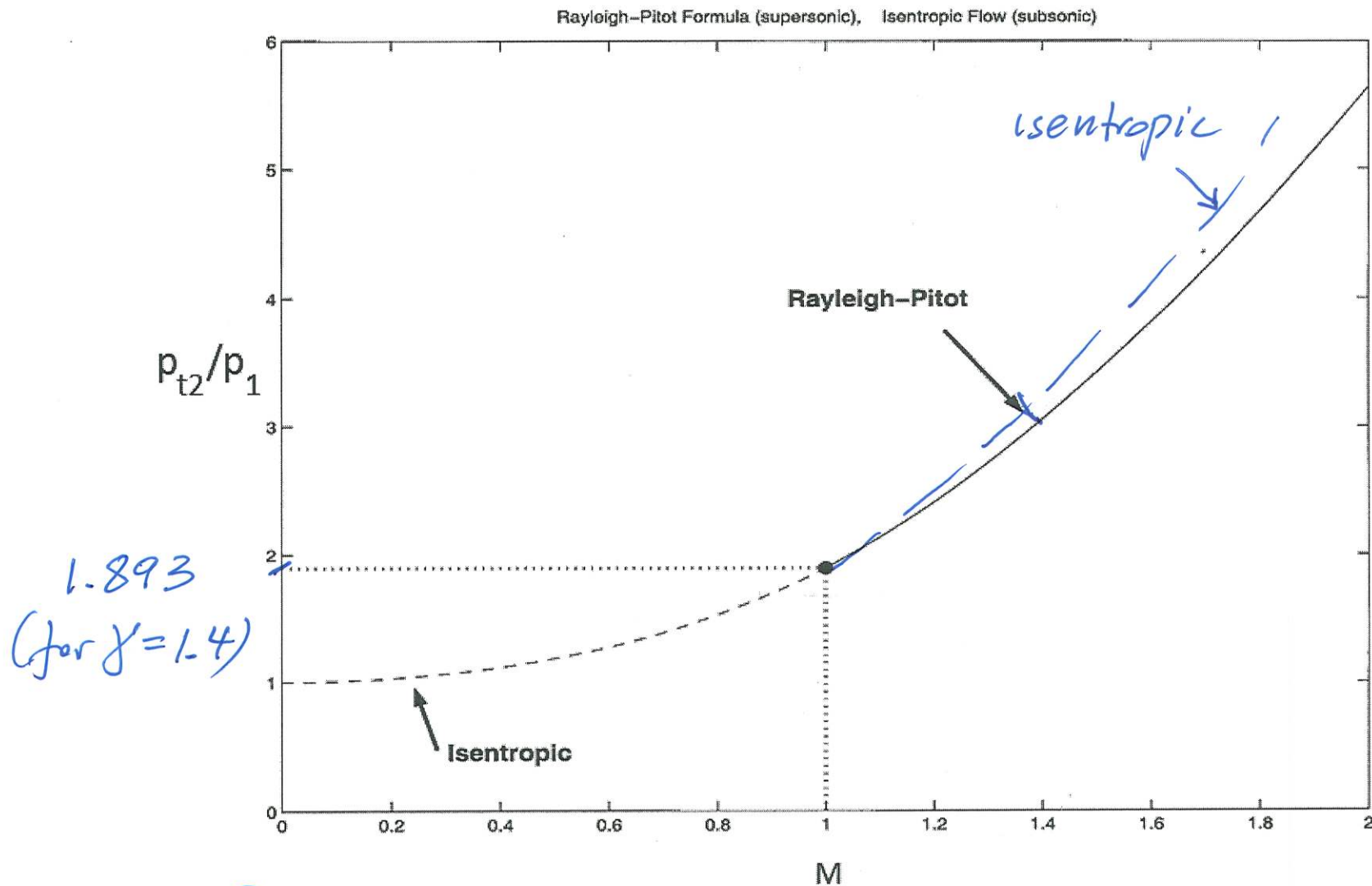
$$M_2^2 = \frac{1 + [(\gamma - 1) / 2] M_1^2}{\gamma M_1^2 - (\gamma - 1) / 2}$$

$$\Rightarrow \frac{p_{t2}}{p_1} = \frac{p_{t2}}{p_2} \frac{p_2}{p_1}$$

Substitute:

$$\Rightarrow \frac{p_{t2}}{p_1} = \left(\frac{(\gamma + 1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma - 1)} \right)^{\gamma/(\gamma-1)} \frac{1 - \gamma + 2\gamma M_1^2}{\gamma + 1} = f(M_1)$$

Rayleigh-Pitot tube formula - *can be part of shock tables*



given $\frac{P_{t2}}{P_1}$ check if $>$ or $<$ than 1.893

$\searrow$ R-P formula ($M > 1$) $\swarrow$ isentropic ($M < 1$)

Example: A Pitot tube is placed in an air stream. The freestream pressure is $P_\infty = 1 \text{ atm} = 101,325 \text{ Pa}$. If the Pitot tube measures a value of stagnation pressure of $1,267,000 \text{ Pa}$, determine the free stream Mach number, M_∞ .

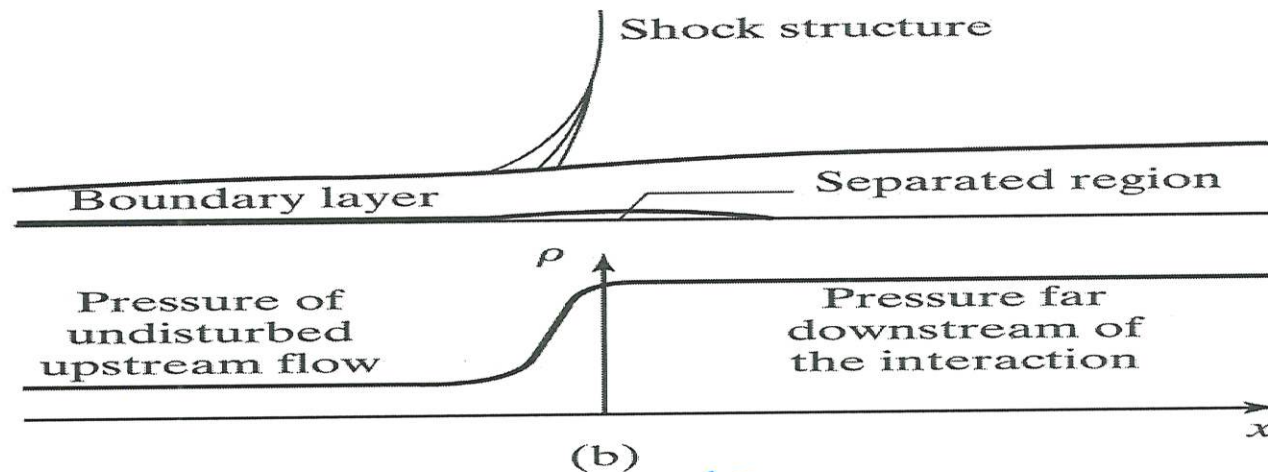
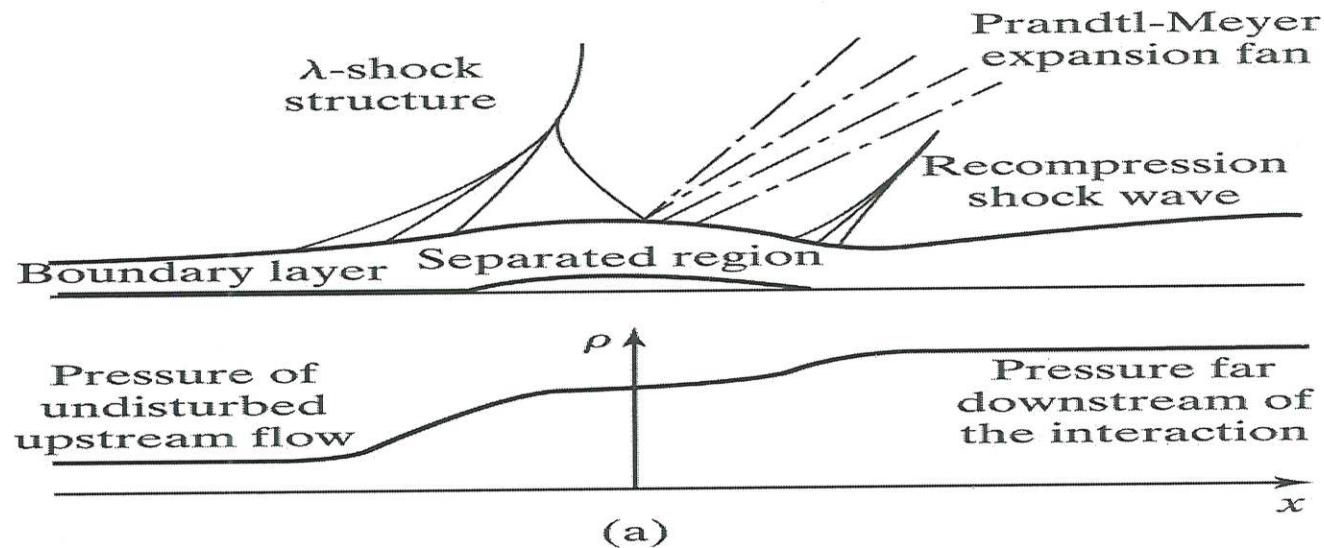
$$\frac{P_{t2}}{P_1} = \frac{1,267,000}{101,325} = 12.5043 \quad (> 1.893)$$

$$P-R \text{ formula: } \leadsto M_1 = 3.05$$

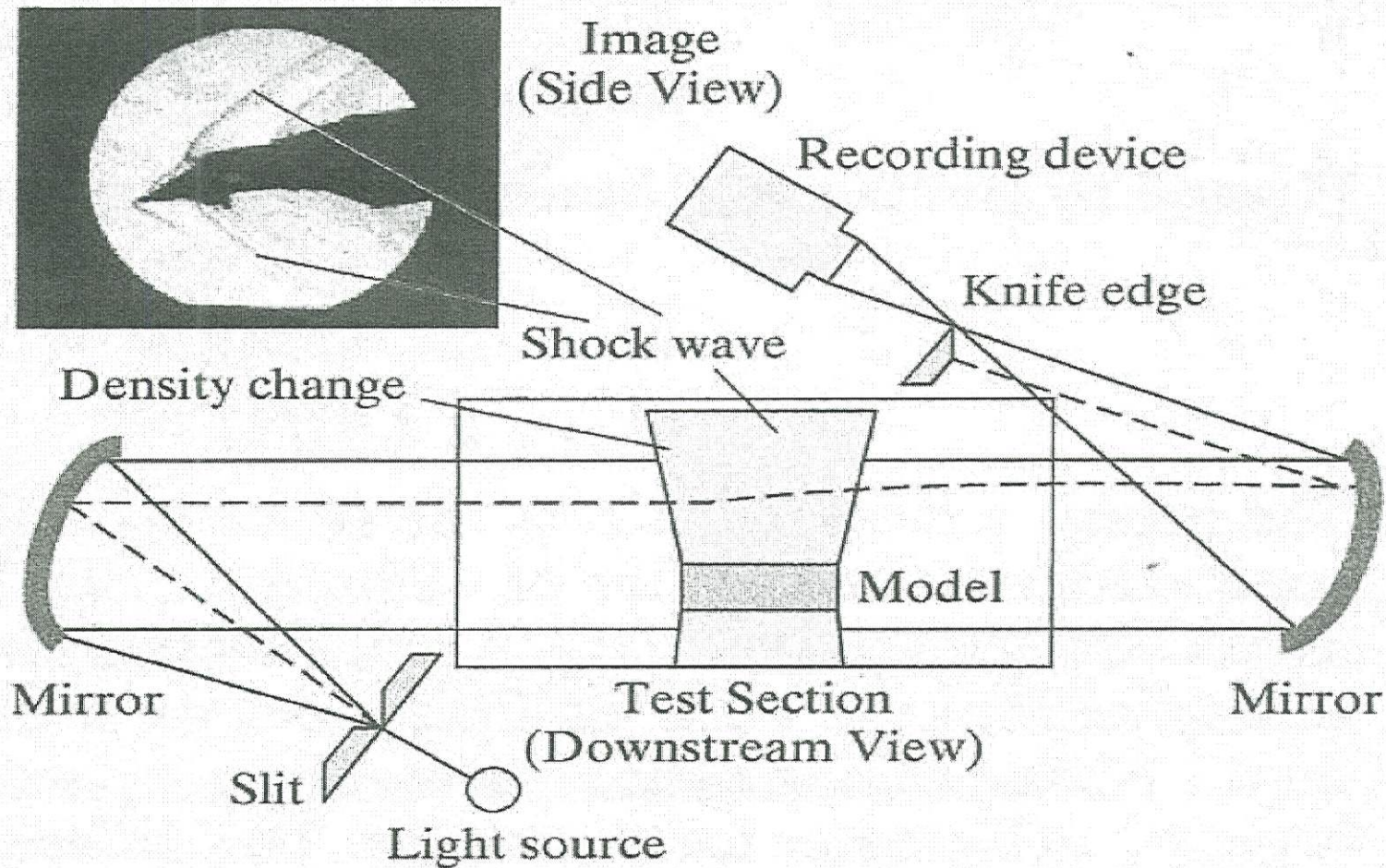
$$\left[\text{note: for isentropic flow} = \frac{P_{01}}{P_1} = 12.5 \leadsto M = 2.3 \right]$$

wrong answer!

Viscous effects (8.7); shock boundary layer interactions



the shock pressure gradient ($\frac{P_2}{P_1} > 1$) may cause separation
(Strong) for a strong shock

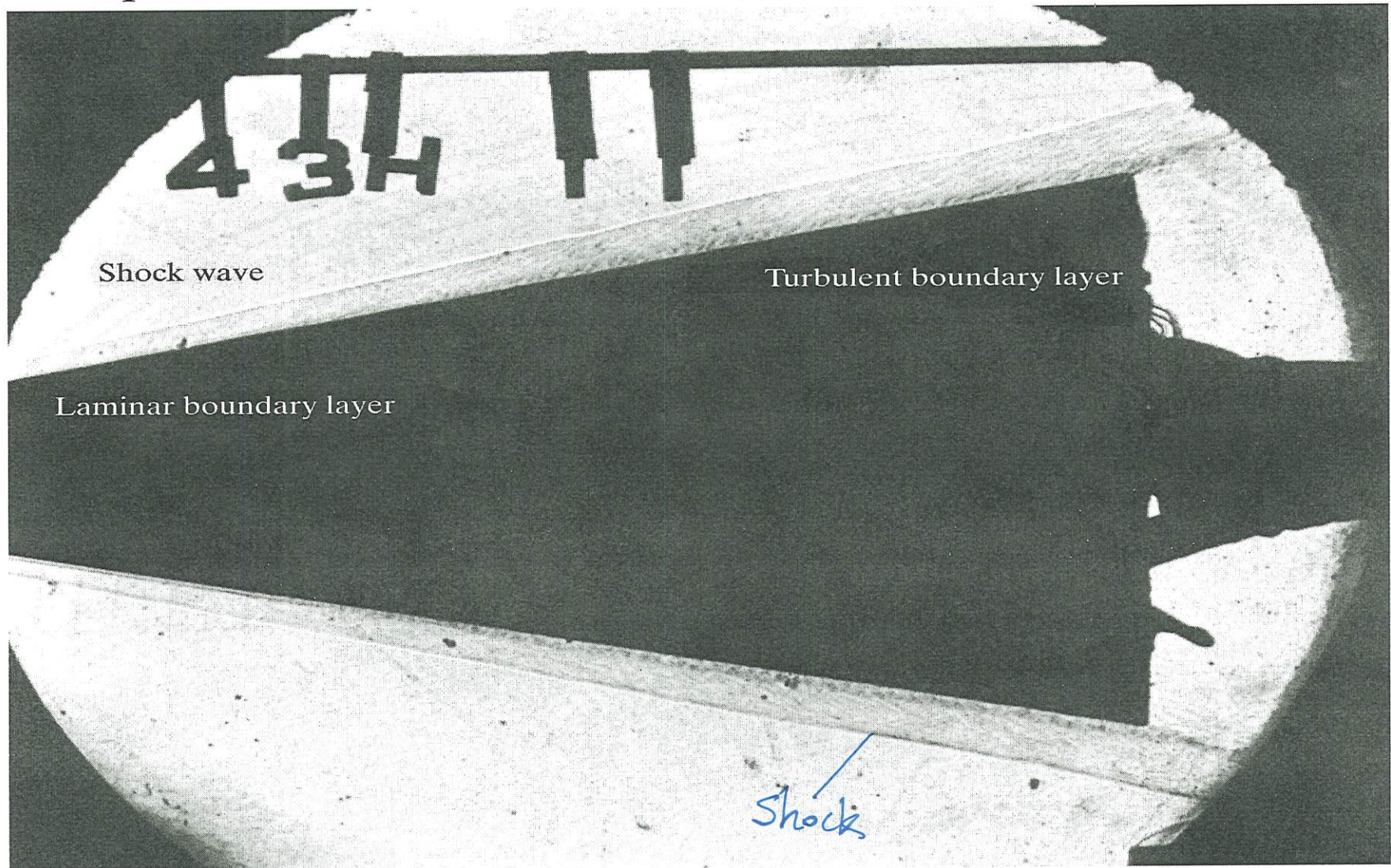


Layout of a Schlieren system
(from NASA Glenn Research Center)

system to detect pressure gradients (i.e. shocks)
- light deflected (bends) misses knife edge - dark lines

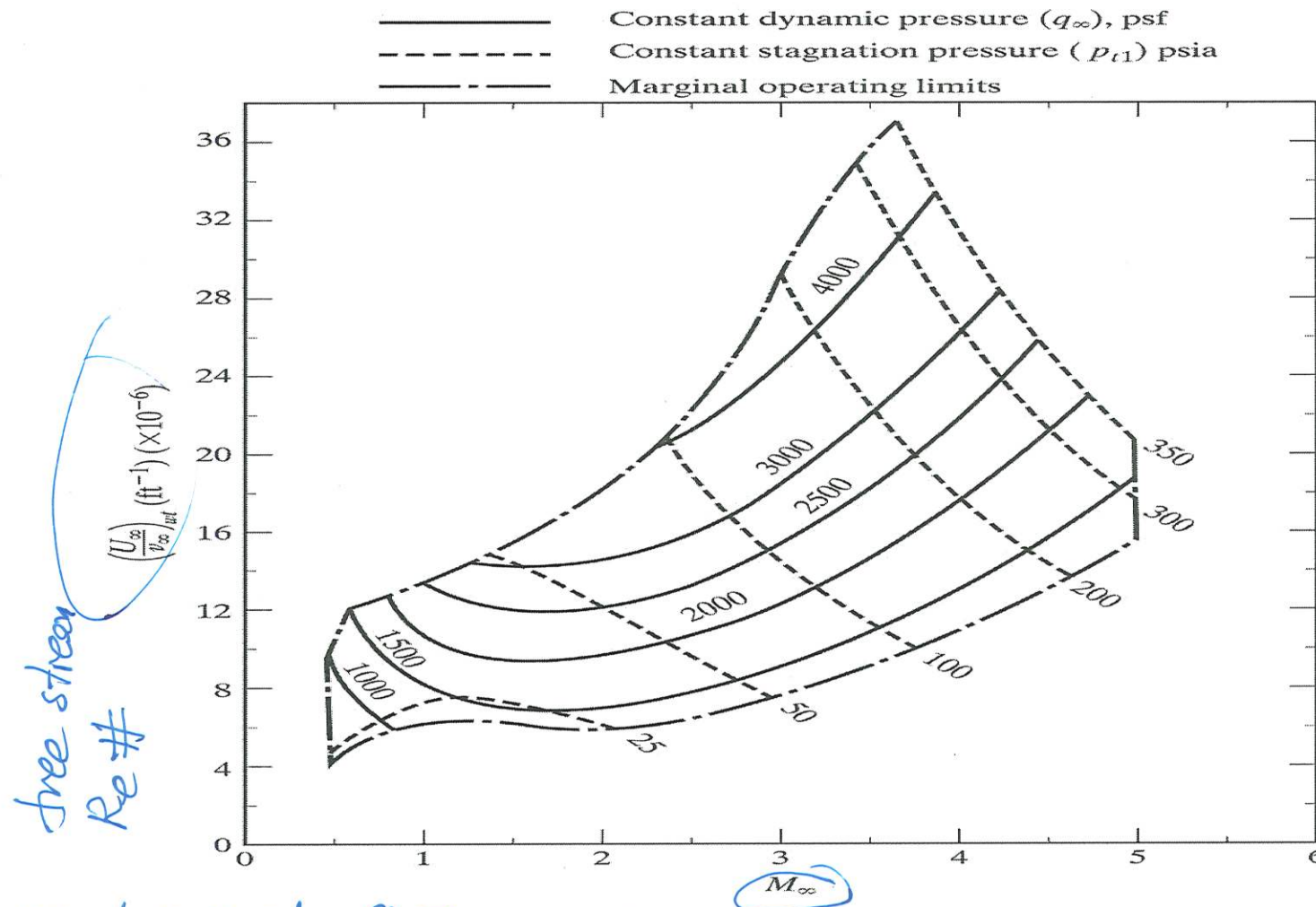
hypersonic

Flow past a slender cone: $M_\infty = 11.5$; $Re_{\infty,L} = 4.28 \times 10^6$; $\delta_c = 12^\circ$.



(operation limits)

Supersonic Wind Tunnels (8.10) Free-stream unit Reynolds number wrt M_∞ for $T_t = 100^\circ\text{F}$ (Vought High-Speed Wind Tunnel)



— read example 8.7: need to match M_∞
 use correlations for $Re_\#$
 note cold T_s